HANDOUTS

PROTECTION, SUPPORT AND MOVEMENT

1.	**Introduction to Protection, Support and Movement** Definition of Animal: A thing having all the properties of life i.e. Nutrition and digestion respiration, excretion, body fluid regulation, reproduction and development etc
2.	**Evolutionary Concept:** Comparative perspective: It is based on the idea that a system cannot be fully understood without comparing it with the systems in different animals.
3.	**Protection: Integumentary system** In Protozoa In Multicellular Invertebrates Animals In Vertebrates Support: Skeletal system Hydrostatic skeleton Exoskeleton Endoskeleton Movement: Non Muscular and Muscular System An Introduction to Animal Muscles Muscular System of Invertebrates Muscular System of Vertebrates
4.	**Integumentary System of Invertebrates (Protozoa)** Integument is the external covering of the animal. It protects the animals from mechanical and chemical injury. It protects the animal from the invasion of microbes. It also regulates the body temperature and excretion of wastes. Also responsible for the conversion of sunlight into Vitamin D

	Plasma Membrane Pellicle Calcium Carbonate shell
5.	**Integumentary System of Multicellular Invertebrates** Epidermis: single layer of columnar cells. Cuticle: may be thin or elastic as in rotifers or thick and rigid as in crustaceans and insects. Calcium Carbonate Shell: as in Cnidarians (corals) Tegument: a complex syncytium- found in parasitic flukes and tapeworms. Exoskeleton: most complex integument found in Arthropods
6.	**Integumentary system of vertebrates (Jawless fishes, Cartilaginous fishes)** Jawless fishes include Lamprays and Hagfishes. Of several types of epidermal glands, one secretes cuticle. In Hagfishes, slime is secreted by multicellular epidermal glands and covers the body surface. Skin of Cartilaginous fishes (sharks) is multilayered and contains mucous cells and sensory cells. Dermis contains small placoid scales in the form of denticles similar to vertebrate teeth. The denticles are protective in function.
7.	**Integumentary system of bony fishes** Bony fishes (teleosts) contain dermal scales. The dermal tissue is covered over by epidermis. The scales grow at the margins and at the lower surface. Growth lines thus developed are useful to determine the age of the fish. The skin of the bony fishes is permeable to gases. The dermis is richly supplied with blood to facilitate in respiration. The epidermis also contains mucous glands. Mucus prevents the fish from infection.

	In some species the mucus contains poisonous alkaloids. Deep water fishes have rhodophores that help lures and warnings
8.	**Skin of Amphibians, Skin of Reptiles** Amphibian skin has layers i) Epidermis ii) Dermis Epidermis is stratified. Dermis is thick containing, pigment layer, mucous & serous glands. During evolution keratin production increased in outer layers of the skin. Skin of Reptiles: i) Epidermis is thick, outer most layer is stratum corneum. Epidermis is modified into keratinized structures e.g. scales, scutes, rattles, claws plaques and spiny crests. These structures are protective in function. Skin of Reptiles: Molting or shedding of skin also occurs in reptiles e.g. snakes. The skin of reptiles reflects their greater commitment to a terrestrial existence.
9.	**Skin of Birds** Skin of birds show many reptilian features with no epidermal glands except uropygial or preening gland. Epidermis is thin having two to three cell layers. Outer keratinized layer is quite soft. Feathers are the most prominent epidermal derivatives. The dermis of birds is similar to those of reptiles containing blood and lymphatic vessels, nerves and sensory bodies. Feathers are important in thermal regulation, flying and behavior.

10. **Skin of Mammals I**

Notable features of mammalian skin are:

1. Epidermis
2. Epidermal glands.
3. Hair
4. Dermis

Rapid cell division occurs in the deeper parts of epidermis.

As these cells come up the surface, where they die and become keratinized.

Later these cells become the outer skin layer, and called the stratum corneum (SC).

The SC is the first line of defense against toxins and microbes.

Dermis is the thicker portion of mammalian skin.

It contains blood vessels, lymphatic vessels, nerve endings, hair follicles, small muscles and glands.

The Hypodermis lies underneath the dermis that consists of loose connective tissue, adipose tissue, and skeletal muscles.

Main Functions of Mammalian Skin:

Main barrier of the body

Regulates body temperature by sweating.

On exposure to sunlight a chemical present in the skin is converted into Vitamin D.

Skin is an important sense organ containing sensory receptors for heat, cold, pressure and pain.

Mammalian Skin Glands:

Sudoriferous glands (sweat glands)

Sebaceous glands secrete sebum

Sebum is a:

i) Lubricant,

ii) Skin-softening agent and

iii) Also acts as a pheromone.

	Pigmentation of the skin: *Chromatophores*- Melanin in human skin. • Other skin colors in various mammals help camouflaging, communication, reproductive status etc, • Some colors is due to color of blood. • Hair is composed of keratin-filled cells that develop from epidermis. *Arrector pili* muscles are attached to the connective tissue surrounding the hair follicle. • On contraction of pili muscles the hair gets erect. This condition is called "Goose hump" in human. • In other mammals this condition helps warm the animal in cold environment. • Nail, is one of the characters of Pimates. like hair, nails are the modifications of epidermis. • Other mammals have claws and hoofs. • Horns are keratinized derivatives of mammalian skin. (Horns are not to be confused with the bony antlers).
11	**Movement and Support: Skeletal system in invertebrates** Body size increased dramatically as organisms evolved from unicellular to multicellular animals. Systems involved in movement and support evolved simultaneously. With respect to support, organisms have three kind of skeletons: 1. Hydrostatic 2. Exoskeleton 3. Endoskeleton Four cell types contribute to movement: 1. Amoeboid cells 2. Flagellated cells 3. Ciliated cells & 4. Muscle cells

12	**Hydrostatic skeleton** Four cell types contribute to movement: 1. Amoeboid cells 2. Flagellated cells 3. Ciliated cells & 4. Muscle cells Earthworm: Longitudinal and circular muscles contract alternately creating a rhythm that moves the earthworm. Hydro skeleton keeps the body from collapsing when the muscles contract.
13	**Exoskeleton** Exoskeleton is a rigid external frame-work of the body. Functions: • Provides site for muscle attachment. • Supports and protects the body like a shield. • Prevents internal soft tissues from drying out. • Provides protection from enemies. Exoskeleton in Arthropods: • Cuticle waterproofs the body. • It limits the animal growth. • Periodic shedding of exoskeleton. • At the joint regions the cuticle is flexible, where the antagonistic muscles function. • At wing joints, a protein, resilin, stores energy on compression and then releases energy to produce movement. • This is reason for success of arthropods.

14	**Endoskeleton** Introduction: Endoskeleton is the internal framework of the body. Examples: Spicules in Sponges. Calcareous plates (ossicles) in echinoderms Bones in vertebrates.
15	**Mineralized tissues in vertebrates** What are mineralized tissues? Tissues in which inorganic calcium carbonate crystals are embedded in the collagen matrix. About two-third of the living species that contain mineralized tissues are invertebrates. Examples: • Gastropods • Lower chordates
16	**Skeletal System of Vertebrates: Cartilage, Bone.** Introduction: Skeletal System of Vertebrates is an endoskeleton enclosed by other tissues. The endoskeleton consists of cartilage and bone. <u>Cartilage is a specialized type of connective tissue that provides:</u> • Support • Helps in movement at joint • Site for muscle attachment. Histologically it consists of: 1. Cells (chondrocytes) 2. Fibers 3. Cellular matrix

	Bone or Osseous tissue has the following functions. • Provides a point of attachment for muscles. • Support the internal organs of the animal. • Store reserves calcium and phosphate. • Manufactures blood cells. • Bone cells also called osteocytes are located in minute chambers called lacunae, which are arranged in concentric rings around the osteonic canals. • These cells communicate with nearby cells through small channels called canaliculae.
17	**Skeleton of fishes** Bone cells also called osteocytes are located in minute chambers called lacunae, which are arranged in concentric rings around the osteonic canals. These cells communicate with near by cells through small channels called canaliculae.
18	**Skeleton of Tetrapod and Human** Axial Skeleton: • Skull • Vertebral Column • Ribs Appendicular Skeleton • Appendages (Fore and hind limbs) • Girdles Pectoral and Pelvic) Skeleton of Tetrapods Amphibians needed support to replace buoyancy. The adaptations for support on land are: Intervertebral discs Rigid bony skeleton

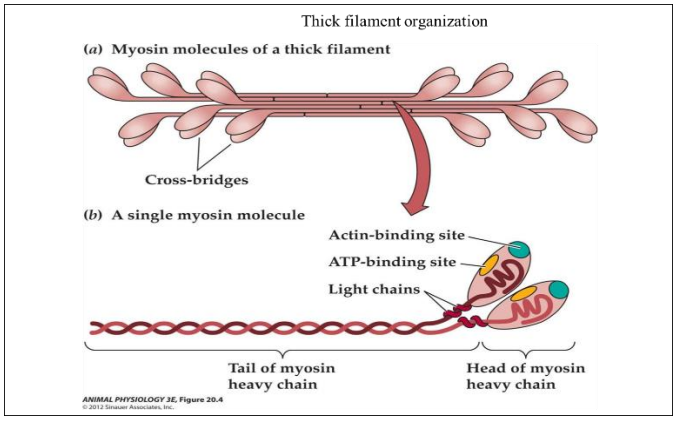
19	**Movement:** Nonmuscular and Muscular Systems Difference between Movement and Locomotion: Movement is the act of changing location from one place to another e.g. a trolley, a ship Locomotion is the movement of an organism by its natural means. e.g. fish, insect, bird Movement is the characteristic of certain cells, protists and animals: For example: certain WBCs coelomic cells and protists. Movement is also brought about by flagella, cilia and pseudopodia. Movement in invertebrates and vertebrates is due to the muscles and muscular systems.
20	**Non Musclar movements** Protozoans move by means of : Pseudopodia, Flagella and Cilia In amoeboid movement, the fluid endoplasm flows forward into the fountain zone of an advancing pseudopodium. As it reaches the tip of pseudopodium endoplasm changes into the ectoplasm. At the same time the ectoplasm near the opposite end in the recruitment zone changes into endoplasm and begins flowing forward. Flagellar and Ciliary Movement: • Structurally, cilia and flagella are similar. • The difference is that cilia are shorter and numerous where as flagella are long and occur singly or in pairs • Cilia and flagella occur in every animal phyla except Arthropoda. • Ciliary movements are well coordinated, for example, in Protozoa. • Cilia occur in rows.

	• Rows of cilia beat out of phase with one another with the result waves periodically pass over the surface. • Direction of movement is brought about by beating the cilia in reverse direction.
21	**Introduction to Animal Muscles** Muscle is a contractile tissue, having cells the muscle fibers. Physiological properties of muscle tissue are: Contractility or elasticity Extensibility Excitability or irritability Types of muscle tissue: Smooth Skeletal and Cardiac <u>Smooth Muscles:</u> Smooth muscles are involuntary. They are uni-nucleated & spindle shaped. They contract slowly and sustain prolonged contraction and do not fatigue easily. Can maintain good tone without nervous stimulation. Smooth muscles are predominant muscle type in invertebrates. Example is adductor muscles in clams and bivalves. <u>Cardiac Muscles:</u> Cardiac muscles fibers are involuntary. Striated and have single nucleus. They are branched. This branching allows the fibers to interlock for greater strength during contraction. They do not fatigue because relax between contractions <u>Skeletal Muscles:</u>

	SM are voluntary. Skeletal muscle cells or fibers are multinucleated and striated. These muscles are associated with skeleton. The skeletal muscles work in antagonistic pairs.
22	**Muscular system of invertebrates** Pedal Locomotion: It is the movement by means of waves of activity in muscular system It is a type of locomotion that occurs in flatworms, some cnidarians and certain gastropods. Accordion-Like Locomotion: Example is earthworm. The longitudinal and circular muscles generate accordion-like waves. Looping Movements: Anterior and posterior suckers in leeches provide alternating points of attachments. Caterpillars exhibit same type of locomotion. Looping Movements: Anterior and posterior suckers in leeches provide alternating points of attachments. Caterpillars exhibit same type of locomotion. Movement by Tube Feet Water vascular system in echinoderms is a unique means for locomotion. Along each canal there are reservoir ampullae and tube feet. Water is driven into the tube feet after passing through the ampullae
23	**Terrestrial locomotion in Invertebrates** Walking: The elements required for walking are: Flexible joints Tendons and Muscles that attach to the exoskeleton.

	Walking limbs of highly evolved arthropods are uniform in structure. Limbs: These are composed of series of jointed elements that become less massive toward the tip. Each joint is articulated to allow movement in only one plane i.e. flexion and extension. Jumping: To jump, the insect exerts a force against the ground to take off with a velocity greater than weight. Jumping insects have relatively long legs having femur, tibia and tarsus. When a flea is resting, the femur of the leg is raised, joints are locked and energy is stored in the protein, resilin. As it to begins to jump joints are unlocked. The force exerted against the ground by the tibia gives the flea a jump. The jump is the result of explosive release of the energy stored in the resilin. The legs with cuticle act as levers in this system. For example: Flea, Grasshopper etc. Flight: Among insects there are two mechanisms of flight: i) Synchronous ii) Asynchronous In synchronous flight the muscles at the base of the wings cause the upward and downward thrust by their contraction and relaxation. This depends upon the nerve impulse. In synchronous flight the muscles at the base of the wings cause the upward and downward thrust by their contraction and relaxation.
24	**Muscular system of vertebrates** In vertebrates, Locomotion occurs by the combined association of: Endoskeleton, Skeletal muscles,

	Tendons Tendons are tough fibrous bands that attach muscles to the skeleton.
25	**Fish musculature** • In fish the musculature consists of segmental myomeres. • The myomeres are arranged along the vertebral column in 3D 'W' shaped blocks separated by collagenous sheaths called myosepta. • Embedded in the myosepta are distinct tendons. • Within teleosts these tendons often ossify. • These ossified myoseptal tendons are homologous to inter muscular bones. • These intermuscular bones are of various shapes, some are Y shapes, some are brush like etc. • Fish movements are based on myomere contraction. • These myomeres cause the lateral undulations of the trunk and the tail. • These undulations bring about fish locomotion.
26	**Structure of Skeletal muscle I** Skeletal muscle consists of a bundle of long fibers running the length of muscle. Each muscle fiber is a single cell with many nuclei below the cell membrane, the sarcolemma. This reflects its formation by the fusion of many embryonic cells. Each muscle fiber is surrounded by connective tissue called <u>Endomycium.</u> <u>Perimycium</u> is connective tissue that wraps bundles of muscle fibers. <u>Epimycium</u> wraps the whole muscle. Each muscle fiber is itself composed of smaller myofibrils arranged longitudinally. The myofibrils, in turn , are composed of myofilaments, which are i) Thin and ii) Thick Each muscle fiber is itself composed of smaller myofibrils arranged longitudinally.

	Skeletal muscle is also called striated muscle because of repeating pattern of light and dark bands. Each repeating unit is called a sarcomere. Sarcomere is a basic functional unit of the muscle. The borders of sarcomere is called Z-line. The Z-lines of adjacent myofibrils contribute to the striations visible with light microscope. Thin filaments are attached to Z lines and project towards the center of the sarcomere They are 7-8 nm in dia. Thick filaments are centered in the sarcomere and are 16nm in diameter. 
27	**Structure of Skeletal Muscle II** Electron Miicroscope reveals the placement of muscle proteins-actin and myosin within the sarcomere. The EM also reveals the detail of A and I bands in the myofibril. A-band is the broad region that corresponds to length of thick filament. I-band corresponds to the thin or actin filaments. H-Zone is in the center of A-band and contains only thick or myosin filaments. Arrangement of thick and thin filaments: Each thick filament is surrounded by six thin filaments. Ultra structure of muscle proteins:

	The actin molecules are arranged in two chains, which twist around each other. Twisting around the actin chains are two strands of other protein, tropomyosin. Another protein in thin filament is the troponin. The thick filament which is about 16 nm in diameter is composed of myosin molecules. Each myosin molecule has a tail terminating in two globular heads. The myosin tail consists of two polypeptide chains coiled together. The heads are called cross bridges.
28	**Sliding Filament Model of Muscle** T –System: Each muscle fiber is surrounded by membrane called sarcolemma. Sarcoplasmic reticulum is a system like endoplasmic reticulum around each muscle fibril. SR lacks ribosomes. T –System: Each muscle fiber is surrounded by membrane called sarcolemma. Sarcoplasmic reticulum is a system like endoplasmic reticulum around each muscle fibril. SR lacks ribosomes. The T-tubule and the terminal portion of the adjacent envelop of SR form Triad at regular interval along the length of fibril. The nerve impulse is carried through the T-tubule to the adjacent SR. Myosin contains globular projections that attach to actin at specific active binding sites forming cross bridges. One cross bridges are formed, they exert a force on the actin filament and cause it to move.
29	**Control of Muscle Contraction** When a muscle contracts sarcomere is reduced i.e. Z-lines are brought closer to gather. In a contracted sarcomere A-band does not change in length but the I-band shortens and the H-band disappears.

	The behavior can be explained by the SF Model. According to this model neither the actin nor the myosin filaments change in length, rather they slide past each other. The sliding of the filaments is based on the interaction of the actin and myosin molecules that make up actin and myosin filaments. Cyclic Interaction Between Myosin and Actin in Muscle contraction: To start with, Myosin head is bound to ATP. Myosin head hydrolysis ATP to ADP and Pi (inorganic phosphate) Myosin head binds to Actin forming a Cross bridge. The releasing ADP and Pi relaxes myosin to low energy state, sliding the actin filament. Binding of a new molecule of ATP releases the myosin head. The myosin head then returns to high energy level and begins a new cycle. *Step A* Myosin head attaches to actin. (High energy ADP + P configuration) *Step B:* Power stroke: Myosin head pivots pulling the actin filament toward the center. Myosin head pivots pulling the actin filament toward the center.
30	**Role of Ca^{++} and Regulatory Proteins in Muscle Contraction** Skeletal muscle contracts only when stimulated by a motor neuron. When muscle is at rest, the binding sites on the actin molecule are blocked by the regulatory protein the tropomyosin. Another set of regulatory proteins, the troponin complex, control the position of tropomyosin on the actin filament. For a muscle to contract Ca ions bind to troponin, that causes the whole tropomyosin-troponin complex to change shape and expose the myosin binding site on actin When Ca++ is present, the sliding of thin and thick filaments occurs and the muscle contracts.

	When internal Ca++ concentration falls, the binding sites of actin are covered and contraction stops. Ca++ concentration in the cytoplasm of the muscle cell is regulated by the sarcoplasmic reticulum. Transportation of Ca++ from cytoplasm into the SR is by active transport.
31	**<u>Regulation of Skeletal Muscle Contraction</u>** There may be hundreds of neurons controlling a muscle each with its own pool of muscle fibers. When a motor neuron produces an action potential all the muscle fibers in the motor unit contracts as a group Action potential trigged by the motor neuron sweeps across the muscle fiber and into it along T-tubule starting the movement of Ca++that regulates the muscle activity. The events that regulate the skeletal muscle contraction are electrical, chemical and molecular.
32	**Energy Supply for Muscle.** Immediate source of energy for muscle contraction is ATP. Supply of ATP is maintained by the Aerobic breakdown of glucose in muscle cell. During exercise, when more energy is required, it comes from Creatine Phophate. Sometimes during strenuous exercise ATP requirement is met by anaerobic breakdown of glucose into Lactic acid. Accumulation of Lactic acid cause muscle fatigue. This also represents Oxygen debit. At rest i.e. after exercise 1/5 of Lactic acid is broken aerobically and its energy is used to change the remaining 4/5 Lactic acid into glucose and later into muscle glycogen. Thus glycogen is replenished again.

HAND OUTS

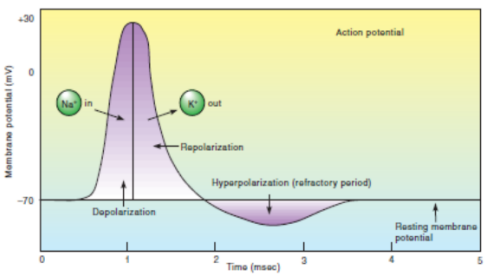
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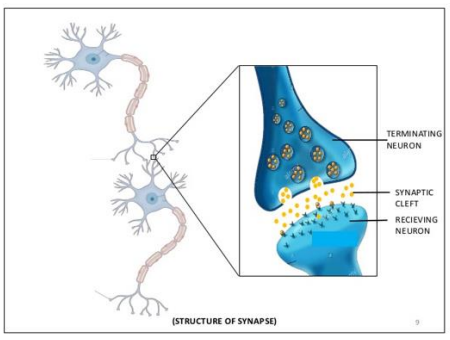
NERVOUS AND SENSORY SYSTEMS

36	**Introduction of NS** Nervous System helps to communicate, integrate and coordinate the functions of various organs and organ systems in animal body. Information flow through NS has three main steps: i) Collection of information from outside and inside the body. (receptors). ii) Processing of information in the NS. iii) Initiation of appropriate response. (effectors). The two forms of communication that integrate body functions to maintain homeostasis are: i) Neurons ii) Hormones. In this chapter i.e. Communication I focuses on the anatomical organization and function of neurons.
37	**Basic Functional Units of NS** Neurons are the basic functional Unit of NS Neurons are specialized cells to produce signals. The signals are communicated to short and long distances. Neurons have Two properties: i) Excitability and ii) Conductivity

	Excitability is the ability to respond. Conductivity is the ability to conduct a signal. Types of Neurons: i) Sensory (receptors) ii) Interneurons (CNS) iii) Motor (effectors). Sensory Neurons act as receptors themselves or activated by receptors. Interneurons comprise the integrating centers. Motor Neurons send the processed information to effectors (muscles or glands)
38	**Neural Pathway between Receptors and Effectors** A stimulus initiates impulse within some sensory structure (receptor). The impulses are then transferred visa sensory neurons to interneurons. After response nerve impulses are generated and transferred via motor neuron to an effector (Muscle/gland) <u>Reflex Action:</u> An action which is carried out at once with the interaction of the will of the animal. or Automatic involuntary motor response. The path of reflex action is the reflex arc.
39,40	**Structure of different kinds of neurons I and II** Neurons are composed of: i) Cell Body ii) Dendrites iii) Axon Cell Body: Central body that contains Nucleus.

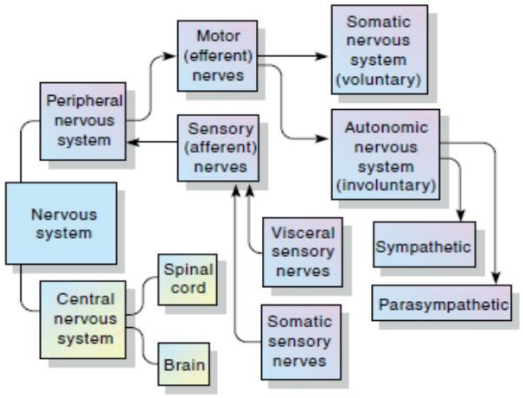
	Dendrites: Short thread like branches extending from cell body. Axon is a long cylindrical process that extends from cell body. Schwann Cell: These are the chain of supporting cells that surround the axon forming insulation layer, the Myelin Sheath. Node of Ranvier: A gap in the myelin sheath between adjacent Schwan Cells. Synapse: Is the junction where the one neuron communicates with the other neuron or muscle or gland cell. Also called Neuronal Junction Synapse: To insulate one neuron from another, and To destroy and remove the carcasses of dead neurons (clean up).
41	**<u>Resting and Action Potential</u>** <u>Resting Potential</u>: The plasma membrane of neuron is polarized, Positive outside and negative inside. R/ potential is measured in mV i.e. 1/1000 volt. Normally R/potential is -70mV. The potential is due to unequal distribution of various electrically charged ions Na+ and K+. Na ions are more outside than inside K ions. Cl ions and negative protein are also more concentrated inside. The pump works to establish the resting potential -70 mV across the membrane. <u>Action Potential</u>: When a threshold stimulus is applied to a point along the resting plasma membrane the permeability of Na+ ions increases at that point. This causes the Resting Potential -70mV towards 0.

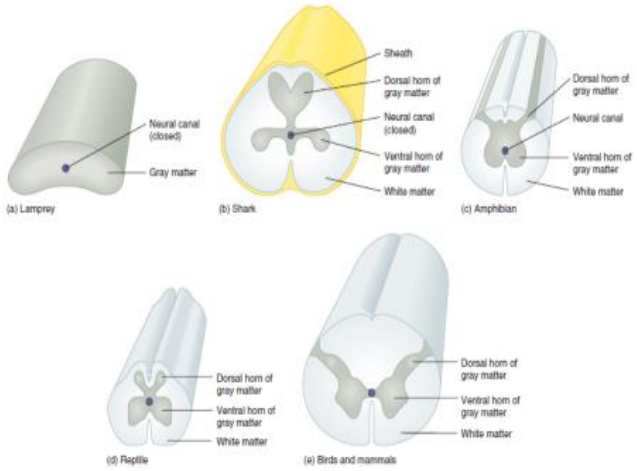
42	**Ion Channels and Sodium Potassium pump** However, the concentration of Na and K ions remain constant on both sides of the membrane. This is due to the action of Na, K, ATPase pump, which is powered by ATP. The pump actively moves 3 Na ions out for each 2 K in. However, the concentration of Na and K ions remain constant on both sides of the membrane. This is due to the action of Na, K, ATPase pump, which is powered by ATP. The pump actively moves 3 Na ions out for each 2 K in. This movement of K+ outside the cell build up the +ve charge again. This causes the membrane repolarized
43	**Nerve impulse (neuron communication)** **It is a wave of electro chemical change.** Or It is the transmission of an action potential along the neuron plasma membrane. It is a wave of depolarization and repolarization. After each A/Potential, there is an interval of time when it is more difficult for another action potential to occur because the membrane has become hyperpolarized (more –ve than -70 mV). This brief period is called refractory period. All or none principle: The principle states that an axon will 'fire' at full power or not at all. Saltatory conduction: Action Potential jump from one node to the next node. This conduction along the myelinated fibers is called Saltatory conduction 

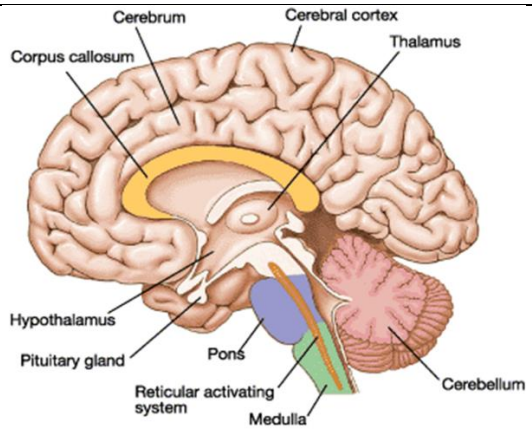
44	**Synapse and its role in Nervous System** <u>Definition of Synapse:</u> It is the junction between two adjacent neurons. There is no cytoplasmic connection between the two neurons. A microscopic gap or synaptic cleft is present between them. The cytoplasmic knob contains numerous mitochondria and small vesicles (50 nm) The vesicles contain neurotransmitter substance Acetyl -choline There is 20 nm gap called synaptic cleft between pre and post synaptic membranes. Transmission of action potential: On reaching the nerve impulse at the Pre synaptic knob, the vesicle release the neurotransmitter into synaptic cleft. The neurotransmitter molecules bind to the receptor on the post synaptic membrane. This triggers action potential in the post synaptic neuron. Presynaptic membrane is always of a neuron but the postsynaptic membrane can be a neuron, muscle or gland. <u>Functions of Synapse:</u> Allow strong signals to pass Block weak signals Select and amplify weak signals. Channel the signals in many directions 

45	**Nervous System of Some Invertebrates I** Nervous system in invertebrate is usually much simpler than the nervous systems found in vertebrates. But there is still a broad range in complexity depending on the type of invertebrate. The simplest type of nervous system is found in hydra and is referred as a "nerve net.“ Nerve nets do not have distinct central or peripheral regions. Lack anything that resembles a brain. Instead, the scattered nerve cells form loose networks in each cell layer of the body wall. Some of these neurons carry information from sensory organs that detect touch, light, or other changes in the environment. Planarians are considered to be among the most primitive animals which acquired the central nervous system (CNS), mesodermal tissues, and bilateral structure during evolution. In Earthworm nervous system can be divided into 2 parts: (i) CNS (ii) PNS CNS is a bilobed brain or cerebral ganglia, sub-pharyngeal ganglia, circum-pharyngeal connectives and a ventral nerve cord.
46	**Nervous System of Invertebrates II** NS in crab differs from that of vertebrates (mammals, birds, fish, etc.) in that it has a dorsal ganglion (brain) and a ventral ganglion. The ventral ganglion provides nerves to each walking leg and all of their sensory organs. Cephalopods have the most highly developed nervous systems among invertebrates.

	Squids have a complex brain in the form of a nerve ring encircling the esophagus, enclosed in a cartilaginous cranium. Neuron size in squid is one meter long and one mm in diameter. Cephalopods have the most highly developed nervous systems among invertebrates. Squids have a complex brain in the form of a nerve ring encircling the esophagus, enclosed in a cartilaginous cranium. Neuron size in squid is one meter long and one mm in diameter. Cephalopods have the most highly developed nervous systems among invertebrates. Squids have a complex brain in the form of a nerve ring encircling the esophagus, enclosed in a cartilaginous cranium. Neuron size in squid is one meter long and one mm in diameter. Ectoneural system: Pertaining to the oral part of the nervous system. Hyponeural system: Part of the nervous system deeper and more weakly developed than the ectoneural system..
47	**CNS in Vertebrates: Basic organization** Characters of Vertebrate Nervous System: Bilateral Symmetry Notochord Tubular Nerve cord Bilateral Symmetry: A body form which is divided into two equal but opposite halves with a central longitudinal plane. Notochord: A rod of mesodermally derive tissue encased in affirm sheath, located ventral to neural tube. First appeared in marine chordates and is present in all vertebrate embryos and is greatly reduced or absent in adults.

	In vertebrates vertebral column replace the notochord. VC led to the development of Tubular Nerve Cord: It is a tube like structure which underwent expansion, modification and specialization into spinal cord and brain. Overtime the anterior end thickened with nervous tissue and divided into fore, mid and hind brain.  NS has two main divisions: CNS (Brain and Spinal cord) PNS (Peripheral NS):All the nerves of the body out side the brain and SC. Nerves are commonly divided into Sensory (afferent) and Motor(Efferent) nerves. NS has two main divisions: CNS (Brain and Spinal cord) PNS (Peripheral NS):All the nerves of the body out side the brain and SC. Nerves are commonly divided into Sensory (afferent) and Motor(Efferent) nerves.
48	**Spinal cords and spinal nerves of vertebrates** <u>Spinal cord:</u> It extends through hollow opening in each vertebra in the vertebral column. In cross section the spinal cord shows neural canal that contains cerebrospinal fluid. Gray matter ,which consists of cell bodies and dendrites. White matter contains nerve processes and axon.

	Extending from the spinal cord are dorsal and ventral roots, which contain sensory and motor fibers respectively. Meninges (sing meninx) surround the spinal cord Dura mater-tough and fiberous Arichnoid- delicate Pia mater-contain blood vessels that nourish the Spinal cord. Dura mater-tough and fiberous Arichnoid- delicate Pia mater-contain blood vessels that nourish the Spinal cord. 
49	**Anatomy of vertebrate brain I** Brain: During development the brain undergoes regional expansion as a hollow tube of nervous tissue. The tube develops into Forebrain, Mid brain and Hind brain.



Forebrain:

The vertebrate fore brain has changed a great deal during evolution.

The forebrain has two main parts- and telencephalon and diencephalon.

Telencephalon, the front part of the fore brain expanded rapidly in size and complexity

Telencephalon consists of :

a. Cerebrum,

b. Olfactory bulb,

c. Limbic system and

d. Corpus Striatum

Cerebrum is divided by a deep groove into Rt and Lt Cerebral hemispheres.

In mammals, the outer most part of cerebrum is called cortex.

Telencephalon consists of :

a. Cerebrum,

b. Olfactory bulb,

c. Limbic system and

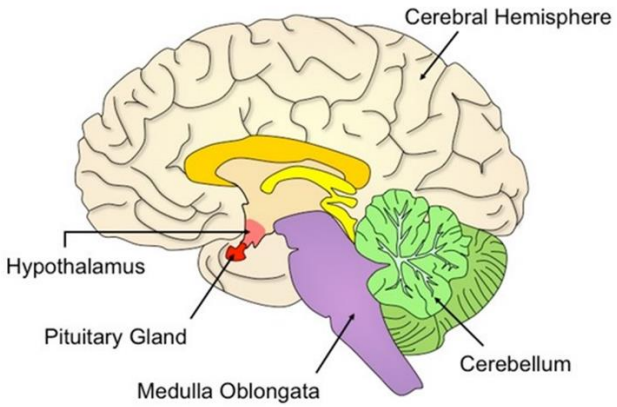
d. Corpus Striatum

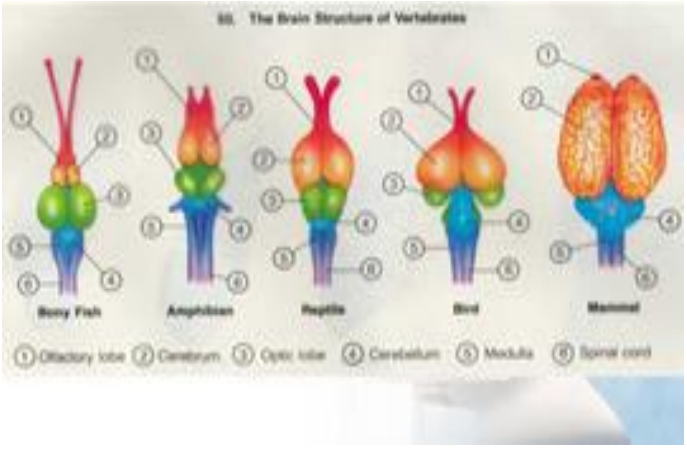
Cerebrum is divided by a deep groove into Rt and Lt Cerebral hemispheres.

In mammals, the outer most part of cerebrum is called cortex.

Telencephalon consists of :

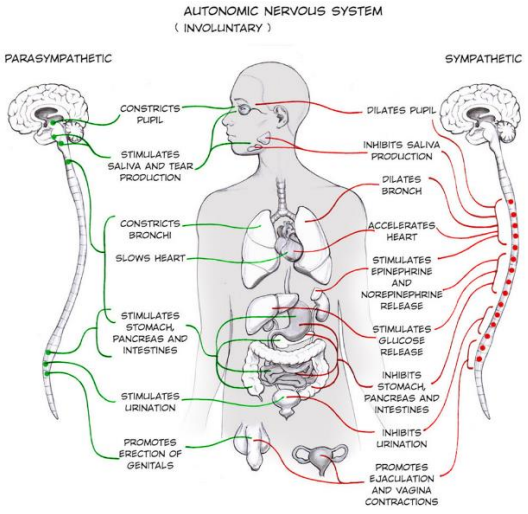
	a. Cerebrum, b. Olfactory bulb, c. Limbic system and d. Corpus Striatum Cerebrum is divided by a deep groove into Rt and Lt Cerebral hemispheres. In mammals, the outer most part of cerebrum is called cortex. The diencephalon expanded slowly as compared to telencephalon. Diencephalon contains-thalamus. Hypothalamus, Pineal gland and Pituitatory gland, The thalamus relays all the sensory information to higher brain centers. The hypothalamus regulates many functions, sexual drive, Carbohydrate metabolism, hunger and thirst. Pituitary gland is a master endocrine gland and produce about 9 hormones.
50	**Anatomy of vertebrate brain II** Mid Brain: Mid brain contains reticular formation, which is a relay center connecting hind brain with the fore brain. Mid brain did not change in size. The roof of the mid brain is a thickened region of grey mater that integrates visual and auditory signals. Hind Brain: Hind brain is continuous with the spinal cord and includes: a. Pons , b. Cerebellum and c. Medulla oblongata Pons is a bridge of transverse nerve tracts from cerebrum to cerebellum. Cerebellum: It is an outgrowth of medulla oblongata. In tetrapods the cerebellum is laterally expanded.

	They provide locomotor control of muscles of appendages. Cerebellum is much larger in birds and mammals. Medulla oblongata: MO is the enlargement where the spinal cord enters the brain. It contains reflex centers for breathing, swallowing, cardiovascular . It is well developed in jawed vertebrates. 
51	**Vertebrate brains (a comparison)** Evolution of Forebrain: The forebrain expanded rapidly in both size and complexity. Many function shifted from the optic tectum to the expanding cerebrum. The increasing importance of the cerebrum affected many brain regions especially thalamus and cerebellum. In mammals outermost part of cerebrum progressively increased in size and complexity. Evolution of Cerebellum: In cartilaginous fishes the cerebellum has distinct anterior and posterior lobes. In teleosts the cerebellum is large in size in active swimmers and small in relatively inactive fishes. In tetra pods the cerebellum is relatively expanded, that provide locomotor control of appendages muscles. In birds and mammals. the cerebellum is much larger. This is reflection of complex locomotor pattern of limb development.

	
52	**Cranial nerves I** <u>I. Olfactory nerve:</u> Olfactory Nerve is the first cranial nerve and conveys special sensory information related to smell. It is the shortest of the cranial nerves and passes from its receptors in the nasal mucosa to the forebrain. <u>II. Optic nerve:</u> It transmits visual information from the retina to the brain. <u>III. Occulomotor:</u> It enters the orbit and innervates muscles that enable most movements of the eye and that raise the eyelid. <u>IV Trochlear:</u> It innervates only single superior oblique muscle of the eye. Controls the downward movement of the eye ball. <u>V Trigeminal:</u> A nerve responsible for sensation in the face and motor functions such as biting and chewing. Its three major branches are: a. Ophthalmic nerve b. Maxillary and c. Mandibular

	V Trigeminal: A nerve responsible for sensation in the face and motor functions such as biting and chewing. Its three major branches are: a. Ophthalmic nerve b. Maxillary and c. Mandibular Mandibular division Supplies Scalp, skin of jaw, lower teeth, lower gum and lower lip. VI Abducens: It is motor nerve. Supplies jaws, floor of mouth and eye muscles. Olfactory and optic are sensory nerves. Occulomotor and Trochlear are motor. Trigeminal is mixed and Abducens is motor nerve.
53	**Cranial nerves II** VII Facial: It emerges from Pons of the brainstem. Controls the muscles of facial expression, taste receptors of anterior 2/3 of tongue, tear glands and salivary glands. Inflammation or damage of this nerve cause Bell's Palsy VIII Vestibulocochlear: It has two branches -Vestibular and Cochlear. The vestibular nerve innervates the vestibular system of the ear. It is responsible for equilibrium. Cochlear supplies the inner ear and serves the sense of hearing. XI Accessary Nerve:

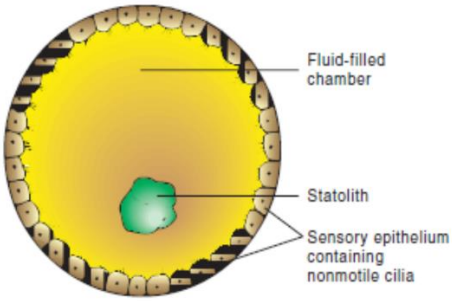
	Has two branches- a) Cranial branch b) Spinal branch Innervates soft palate, pharynx and larynx. It is motor nerve. <u>XII Hypoglossal:</u> It is a motor nerve. Innervates tongue muscles. Facial nerve is a mixed nerve and Vestibulocochlear is a sensory nerve. Glossopharyngeal and Vagus nerves are mixed nerves. Accessory and Hypoglossal are motor nerves.
54	**Introduction of Autonomic NS** Autonomic nervous system (ANS) is the part of peripheral nervous system. PNS also includes Somatic Nervous System (SNS). The SNS consists of motor neurons that stimulate skeletal muscles. In contrast, the ANS consists of motor neurons that control smooth muscles, cardiac muscles and glands. In addition, the ANS monitors visceral organs and blood vessels. In the ANS, the connection between the CNS and its effector consists of two neurons—the preganglionic neuron and the postganglionic neuron. The synapse between these two neurons lies outside the CNS, in an autonomic ganglion. The ANS is further divided into the sympathetic nervous system and the parasympathetic nervous system.
54	**Sympathetic and Parasympathetic NS** Autonomic NS is divided into divisions: Sympathetic and Parasympathetic NS.

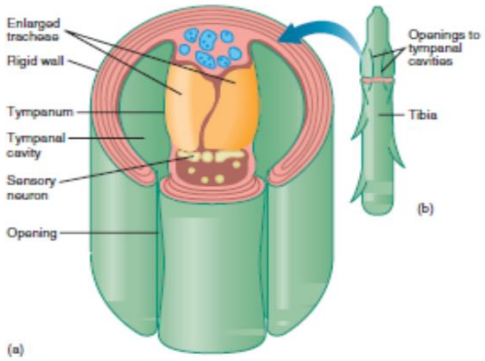
	These two divisions generally make synaptic contacts with the same organ but usually produce opposite effects. Sympathetic NS: It consists of ganglia, nerves and plexues that supply involuntary muscles. This NS arise from middle portion of spinal cord and terminate in ganglia. This system is important during emergency situation. 'Fight and flight" However neither kind of nerve is exclusively excitatory or inhibitory, For example, the sympathetic fibers increase heart beat but inhibit intestinal peristaltic movements. Parasympathetic NS: This system consists of nerves some of which emerge from brain and others from sacral or pelvic region of the spinal cord. PS division is associated with non stressful activities e.g. resting, eating, digestion & urination. Retards heartbeat. 
55,56	**Introduction of sensory reception** In human five senses- sight, smell, hearing, taste and touch are commonly known.

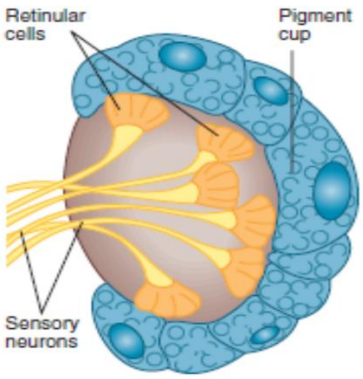
	Apart from these there are other senses also found in animals For example: Invertebrates possess the following sensory receptors: i) Tactile receptors that sense touch. ii) Hygroreceptor that detect content of air. iii) Georeceptors That sense pull of gravity. iv) Proprioceptors- that respond to compression, stretching, bending, and tension. v) Phonoreceptors- That are sensitive to sound vi) Baroreceptors- that respond to pressure changes. vii) Chemoreceptors- that are sensitive to air and water borne chemicals. viii) Photoreceptors-that are sensitive to light. ix) Thermoreceptors- that are sensitive to temp changes. vi) Baroreceptors- that respond to pressure changes. vii) Chemoreceptors- that are sensitive to air and water borne chemicals. viii) Photoreceptors-that are sensitive to light. ix) Thermoreceptors- that are sensitive to temp changes. Sensory receptors convert stimuli into nerve impulse. All receptors are tranducers i.e. that convert one form of energy into another. Different types of receptors convert different kinds of stimuli into local electrical potential called generator potential. When the GP reaches the neuron threshold potential, it causes the channels in plasma membrane to open and creates an action potential. The impulse then travels along the axon towards a synaptic junction and becomes information going to the CNS. The nature of the nerve impulse is the same. How then action potential give rise to different sensations? Animals that have, nerve signals from specific receptors always end up in a specific part of brains for interpretation. For example a stimulus that goes to the optic center is interpreted as visual sensation. And so on.

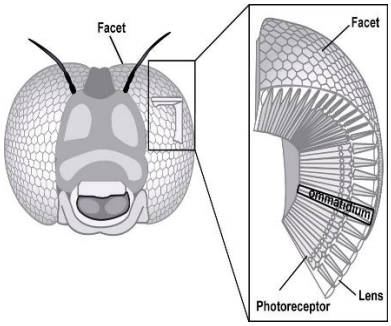
57	**Invertebrate sensory receptors** An animal's behavior is largely a function of its responses to environmental information. Invertebrates possess a variety of receptors through which they receive information about their environment. (a) Baroreceptors (b) Chemoreceptors (c) Georeceptors (d) Hygroreceptors (e) Phonoreceptors (f) Photoreceptors (h) Tactile receptors (i) Thermoreceptors
58	**Baroreceptors** Baroreception A barometer is a scientific instrument used to measure air pressure. Pressure tendency can forecast short term changes in the weather. The Zoologists have not identified any specific structures for baroreception in invertebrates. Nevertheless responses to pressure changes have been identified in ocean swelling copepod crustaceans, Ctenophores, jelly fishes medusa and squids. Some crustaceans that live between the tides respond to water pressure changes and coordinate with daily tidal movements.
59	**Chemoreceptors** Chemoreceptors are sensitive to chemicals. Chemoreception is the oldest and most universal sense in animal kingdom. Examples: Protozoans show avoiding response to acid, alkali and salt stimuli.

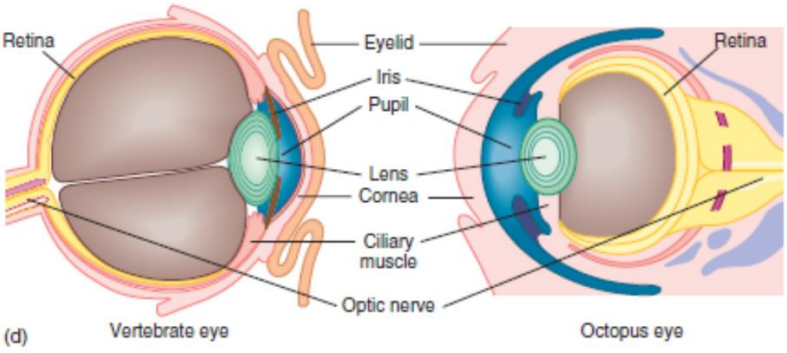
	Specific chemical attract predatory ciliates to their prey Location of chemoreceptors: In aquatic invertebrates they are located in pits or depressions, through which water carrying the specific chemicals may be circulated. In arthrpods, the receptors are usually located on the antennae, mouthparts, and legs in the form of hollow hairs called Sensilla containing chemosensory neurons. The types of chemicals to which invertebrates respond are closely associated with their life styles. For example: chemoreceptors are sensitive to humidity; pH, prey tracking, food recognition, and mate location. With respect to mate location the antennae of male silkworm moth detect one bombykol molecule in over a trillion molecules of air. Female silk moth secrete bombykol as a sex attractant. This enables a male to find a female at night from several miles down wind.
60	**Geo receptors: Statocysts, Statoliths** Georeceptors; Respond to the pull of gravity giving animals information about their orientation relative to up and down. Most georeceptors are statocysts. (standing bladder). Statocyst is a fluid filled chamber lined with cilia bearing sensory epithelium. Within the chamber is a solid granule called statolith. When animal moves, both the movement of statolith and the flow of fluid over the sensory epithelium provide information about the animal's linear and rotational acceleration relative to the environment.

	 Occurrence of georeceptors: They are found in various gastropods, cephalopods, crustaceans, nemertines, polychaetes and scyphozoans. Burrowing animals can not rely on photo for orientation instead they rely on georeceptors within he substratum. Planktonic animals orient in their environment using statocysts. They are important at night and deep waters. Most of aquatic insects detect gravity from air bubbles entrapped in tracheal tubes, which stimulate sensory bristles that line the tubes.
61	**Hygroreceptors** Hygroreceptors are sensitive to water content of air. Some insects have hygrorecptors that can detect small changes in the ambient relative humidity. There hygroreceptors are have been identified on the antennae, palps, underside of the body and near the spiracles. Hygroreceptors enables the insects to seek an environment with a specific humidity to modify their physiology or behavior with respect to humidity. Drosophila detect air humidity through hygroreceptors located in a sac like invagination of the antenna. These receptors rapidly respond to a puff of dry air, potentially alerting the animal to the fact that dangerous dry conditions are looming. Swarming, in termite is humidity dependent.
62	**Phonoreceptors (tympanal organ)** Phonoreceptors are the receptors that respond to sound.

	They have been identified only in insects, archnids and centipedes. Some other invertebrates seem to respond to sound- induced vibrations of the substratum. Crickets, grasshoppers posses phonoreceptors called tympanic or tympanal organs. This organ consists of tough, flexible tympanum that covers internal sac. The sensory neurons are attached to the tympanum.  When sound waves strike the tympanum it vibrates. When sensory neurons are stimulated by these vibrations generator potential is produced. Most arachnids possess phonoreceptors in their cuticle called slit sense organs that are sensitive to vibrations. Centepedes have organs of Tomosvary, which are sensitive to sound. However, the physiology of these organs i.e. the organs of Tomosvary and slit like organs are poorly known.
63	**Photoreceptors (Ocellus) I** Photoreceptors are sensitive to light. All photoreceptors possess light-sensitive pigments e.g. rhodopsin and carotenoids. These pigments absorb photons of light energy and then produce generator potential. This is basic commonality. Photoreceptors in various organisms: <u>Euglena:</u>

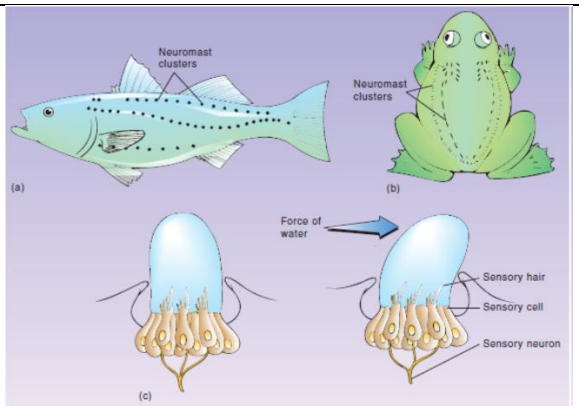
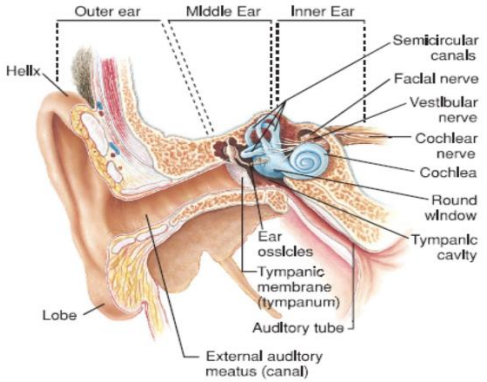
	Stigma (mass of bright red granule containing carotenoid. Actual photoreceptor is the swelling at the base of the flagellum. Stigma serves as a shield, which is essential if the receptor is to detect light. Thus the photoreceptor plus stigma enable euglena to orient itself so that its receptor is exposed to light. This helps the protozoan maintain itself in the region where sufficient light is available. In earthworm *Lumbricus*, simple unicellular photorecepter cells are scattered over the epidermis or concentrated in particular areas of the body. Other animals possess multicellular photoreceptors that can be classified into: ocilli, compound and complex eyes. Ocellus is simply a small cup lined with light sensitive receptors (retilunar cells) and backed by light absorbing pigment. The retilunar cells contain photosensitive pigments. Stimulation by light causes a chemical change in the pigment.  This stimulation leads to generator potential then action potential that sensory neurons carry. This type of visual system gives animal an information about the direction and intensity of light only but no image. Ocelli are common in phyla such as Annelida, Mollusca and Arthropoda.
64	Photoreceptors (compound eye) II

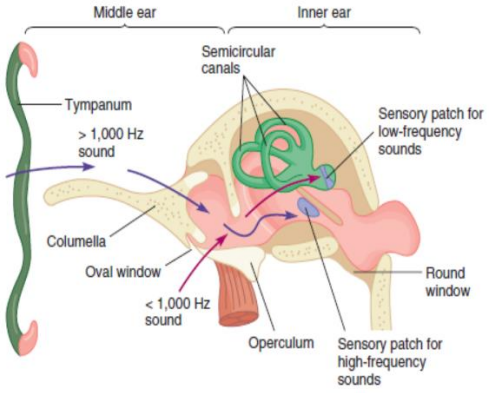
	This stimulation leads to generator potential then action potential that sensory neurons carry. This type of visual system gives animal an information about the direction and intensity of light only but no image.  Compound eyes consist of many clear units called Ommatidia. Compound eyes occur in some annelids and bivalve molluscs. Best developed and understood in arthropods. Compound eyes may contain thousands of ommatidia each oriented differently. The visual field of a compound eye is very wide. Each ommatidium has its own nerve tract leading to large optic nerve. The visual fields of adjacent ommatidia overlap to some degree. The visual field of a compound eye is very wide. Each ommatidium has its own nerve tract leading to large optic nerve. The visual fields of adjacent ommatidia overlap to some degree. Color vision is particularly important in active, day-flying, nectar drinking insects, such as honeybees. Honeybees learn to recognize particular flowers by color, scent, and shape
65	<u>Photoreceptors (camera eyes) III</u> Complex Camera Eyes occur in squids and octopuses.

	These eyes are best image forming eyes among invertebrates. Largest eye size is 25 to 30 cm in diameter 45 ft squid. Cephalopod eye is often compared with vertebrate eye in its structure. The eye is composed of thin transparent cornea and a lens that focuses light on retina. Lens is suspended and controlled by ciliary muscles. The eye of squid, however, differs from vertebrate eye in that the receptor site on the retina face the direction of light entering the eye. In the vertebrate eye the retinal layer is inverted and the receptors are the deepest cells in the retina. In cephalopods light is focused by muscles which move the lens toward or away from the retina and by altering theIn the vertebrate eye the retinal layer is inverted and the receptors are the deepest cells in the retina. In cephalopods light is focused by muscles which move the lens toward or away from the retina and by altering the shape of the eye ball. 
66	**Proprioreceptors** Proprioceptors commonly called stretch receptors. Located internally. Sensitive to mechanically induced changes caused by stretching, bending compression or tension. These receptors give an animal information about the movement of its body parts and their position relative to each other.

	Proprioceptors are associated with appendage joints and body extensor muscles. In these animals the sensory neurons may be attached to the muscles, elastic connective tissue fibers or various membranes that span joints. As shape changes, generator potential starts.
67	**Tactile Receptors** Tactile receptors are generally derived from modifications of epithelial cells associated with sensory neurons. Most tactile receptors of animals involve projections from the body surface. These projections include: bristles, spines, setae and tubercles. When an animal comes in contact with an object in the environment, these receptors are mechanically deformed. These deformations activate the receptors, which in turn activates underlying sensory neuron initiating generator potential. Most tactile receptors are also sensitive to mechanically induced vibrations propagated through water or a solid substrate. Examples: Tube swelling Annelids bear receptors that allow them to retract quickly from their tubes in response to any movement in surroundings. Web-building spiders have tactile receptors that can sense struggling prey in webs through vibrations of the web threads.
68	**Thermoreceptors** Web-building spiders have tactile receptors that can sense struggling prey in webs through vibrations of the web threads. Leeches and ticks possess heat sensing mechanism that can recognize warm blooded hosts. Certain insects, some crustaceans, and the horseshoe crab (Limulus) can also sense thermal variations. In all of these cases, however, specific receptor structures have not been identified.
69	**Vertebrate Sensory Perception** Vertebrate sensory receptors reflect adaptations to the nature of sensory stimuli in environment.

	The environment has chemical and physical characteristics that affect the kinds of energy and molecules that carry sensory information. For example, our external environment consists of the media that surrounds us: the earth that we stand on and the air that we breathe. Other animals may have different external environments: a trout may be immersed in the cool, clear water of a mountain stream. A turtle may be submerged in the turbid water of a swamp; and a salmon may be swimming in the salty water of the sea.
70	**Lateral line system** The lateral line system is electrical sensing that occur both in jawless and jawed fishes and some amphibians, along the sides of head and body It consists of sensory pores in the epidermis of the skin that connect to canals leading into electroreceptors called ampullary organs. These organs can sense feeble electrical field produced by organisms living in surrounding water. This ability to detect these fields help a fish to find mate, capture prey or avoid predators. This is especially valuable sense in deep, turbulent or murky water where vision is of little use. In fact, some fishes actually generate fields and then use their electroreceptors to detect how surrounding objects distort the field. This allows these fishes to navigate in murky or turbulent waters.
71	**Lateral line system & Mechanoreception** Mechanoreceptor also called Neuromasts are the part of Lateral Line system. These receptors are fund in cyclostomes, sharks, aquatic amphibians and some advanced fishes. Neuromasts are located in pits along the body but not in head region. Neuromasts are responsive to local water disturbance.

	 When water near the lateral line moves, water in the pits also moves. This movement disturbs hair like cells of the neuromasts causing the generator potential in the associative sensory neurons. This helps the animal to detect the direction and force of water currents and thus the movement of other animals in the vicinity and become alert from the predators. For example this sense enables a trout to orient the head upstream in water.
72	**Hearing and Equilibrium in air** Hearing is important to vertebrates as a mechanism to: a) become alert against nearby or faraway dangerous activity. b) become important for the location of food communicated by other animals e.g. Crows. c) respond the calls for mating e.g. cats, cows. Hearing and Equilibrium are the sensations that occur together in the same vertebrate organ, ear.  For hearing it is the Auditory organ and for equilibrium and posture, the Vestibular apparatus.

	The main part of auditory apparatus which is involved with hearing is the cochlea. The part of the ear which is involved with equilibrium are the semicircular canals. Sound results when waves transmit energy through some medium such as air or water. Adaptation of hearing in air resulted from the evolution of an acoustic transformer that incorporates a thin membrane that is exposed to air.
73	**Tympanum in Amphibians** Tympanum or the ear drum first evolved in Amphibians. Ears of Anurans consists of : a) Tympanum b) Middle ear and c) Inner ear Tympanum is modified integument stretched over cartilaginous ring.  Touching the tympanum is an ossicle, called columella or stapes. The opposite end of the stapes touches the membrane of the oval window that stretches between middle and inner ear. There are 3 semicircular canals which attached by their end. The semicircular canals are fluid filled. A second small ossicle, the operculum, also touches the oval window. Muscles attached to the operculum and columella can lock either or both of these ossicles, allowing a frog to screen out either high or low frequency sounds. The semicircular canals are fluid filled.

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74	**Hearing in Reptiles** Reptilian ears vary in structure. Ears of snakes lack a middle ear cavity and tympanum. They have vestiges of the hearing apparatus inside their heads. A bone of jaw articulates with the stapes and receives vibrations of the ground or substratum. Reptilian ears vary in structure. Ears of snakes lack a middle ear cavity and tympanum. They have vestiges of the hearing apparatus inside their heads. A bone of jaw articulates with the stapes and receives vibrations of the ground or substratum. Airborne sounds are transmitted from the skin receptors to the lungs and then to the inner ear and 8th cranial nerve to the brain. Snakes hear low frequency sounds. In some reptiles tympanum or ear drum may be on the surface or in small depression in the head. The structure of the inner ear is similar to that of amphibians.
75	**Hearing in Birds** The ear openings in birds are just below and behind the eyes and are covered by fine feathers called auricular or ear coverts. These ear coverts funnel sounds from to the ear opening down to the eardrum. Movement of the fluid caused by the vibration of the membrane moves these hairs, or cilia.

	In birds there is just one stapes or columella, which transmits vibrations from the eardrum to the membrane across the oval window in the cochlea of the inner ear. The cochlea is a hair lined, fluid-filled chamber. The movement of these hairs excites nerve endings to transmit signals to the brain where it is translated and interpreted as sound. There is another opening in the cochlea, the round window, which is covered by yet another membrane and this allows the pressure waves moving the cilia to dissipate. Also in the inner ear are the semicircular canals used to determine orientation and to regulate balance.
76	**Anatomy of Human Ear** In birds there is just one stapes or columella, which transmits vibrations from the eardrum to the membrane across the oval window in the cochlea of the inner ear. The cochlea is a hair lined, fluid-filled chamber. Three small ossicles are the parts of middle ear. These ossicles include: i) malleus (hammer) ii) incus (anvil) iii) stapes (stirrup) The malleus adheres the tympanic membrane and connects to the incus. The incus connects to the stapes which adheres to the oval window. Auditory or eustachean tube extends from the middle ear to the nasopharynx. This equalizes air pressure between middle ear and throat. Auditory or eustachean tube extends from the middle ear to the nasopharynx. This equalizes air pressure between middle ear and throat. The entire inner ear is bathed in a cushioning fluid, the endolymph. The sensory cells which have hair-like projections are called Organs of Corti. These organs are located on the basilar membrane that forms the base of cochlea.
77	**Hearing of Human Ear** Sound waves enter the outer ear and reach the tympanic membrane to vibrate.

These vibrations move the ossicles one after the other against the oval window.

When the middle ear transfers the vibrations to the cochlea the fluid in the cochlea is displaced.

This displacement of the fluid make the hair cells move.

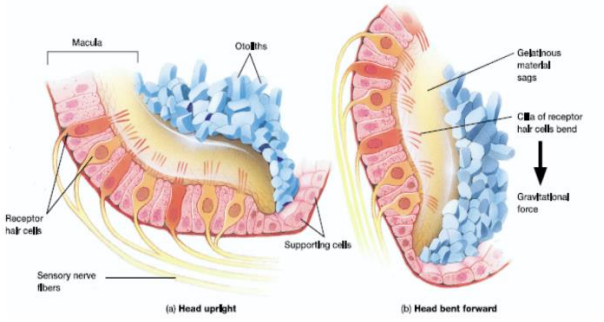
Signals from the cells are converted into nerve impulse and sent to the brain through the auditory nerve, thereby helping the process of hearing.

Receptor cells of the organ of Corti which have hair-like projections bend causing generator potential, which leads to action potential that travels along the vestibule cochlear nerve to the brain for interpretation.

When the body is still, the otoliths in the semilunar canals rest on hair cells.

When the head horizontally or vertically the granules are displaced. This displacement bends the hair slightly so that hair cells initiate generator potential.

Continuous movements cause motion sickness.



The sense of equilibrium can be divided into two equal senses- Static and Dynamic

Static refers to sense movement in one plane (vertical or horizontal).

Dynamic refers to angular or rotation movements.

Humans are not able to hear low-pitched voice below 20 cycles per second.

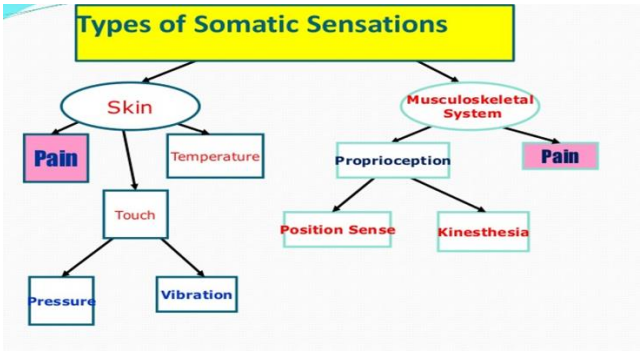
Young children can hear high pitched sound up to 20,000 cycles per second.

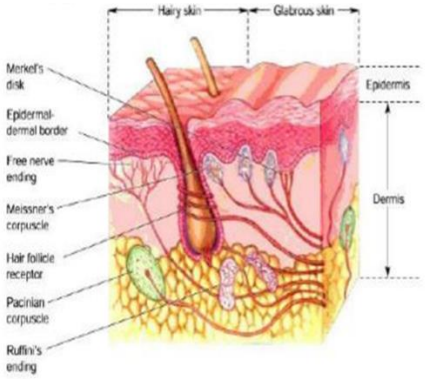
Dog can easily detect sound of 40,000 cycles per second.

An otolith also called statoconium or otoconium or statolith, is a calcium carbonate structure in the saccule or utricle of the inner ear, specifically in the vestibular system of vertebrates.

In mammals, otoliths are mall particles, composed of a combination of a gelatinous matrix and calcium carbonate in the viscous fluid of the saccule and

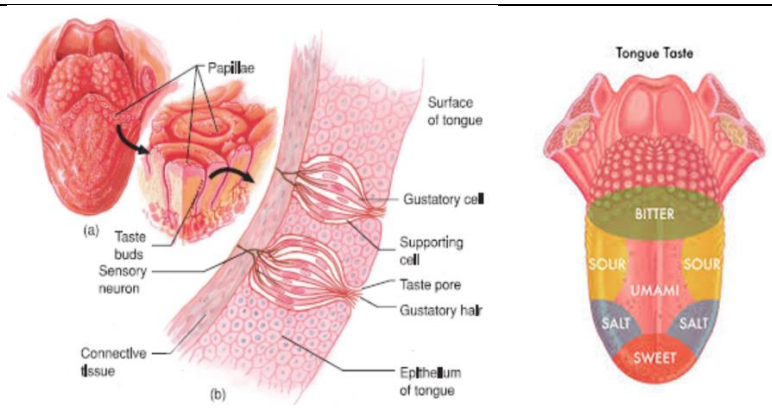
	utricle. The inertia of these small particles causes them to stimulate hair cells when the head moves.
78	**Hearing and Equilibrium in Water** In bony fishes, the receptors for equilibrium and hearing are in the inner ear. Semicircular canals detect the rotational movements by detecting the direction of gravitational pull. Since the fishes lack the outer and middle ears, the vibrations pass from the water through the bones of the skull to the inner ear. A few fishes have chains of bony ossicles that pass between the swim bladder and back of the skull. Sound waves that enter the pharynx are transmitted to swim bladder causing it expand and contract according to the intensity of incoming waves. The vibrations of contraction pass forward along the chain of ossicles and then to the inner ear. Sharks' hearing, have a sharp sense of hearing and can possibly hear prey many miles away in water. Some species of Carp and Herring, hear through their swim bladders, which function like a hearing aid. Fish can also sense sound through their lateral lines and their otoliths (ears).
79	**Skin Sensors of Damaging Stimuli** Pain receptors are bare or naked nerve endings through out the body of mammals except for the brain & intestine. These nerve endings are called Nociceptors. The may be: External nociceptors- found in tissue such as the skin, the corneas, and the mucosa. Damaging or noxious stimulus is actually or potentially tissue damaging event. Noxious stimuli can either be: Mchanical (pinching or other tissue deformation), Chemical (exposure to acid or irritant), or

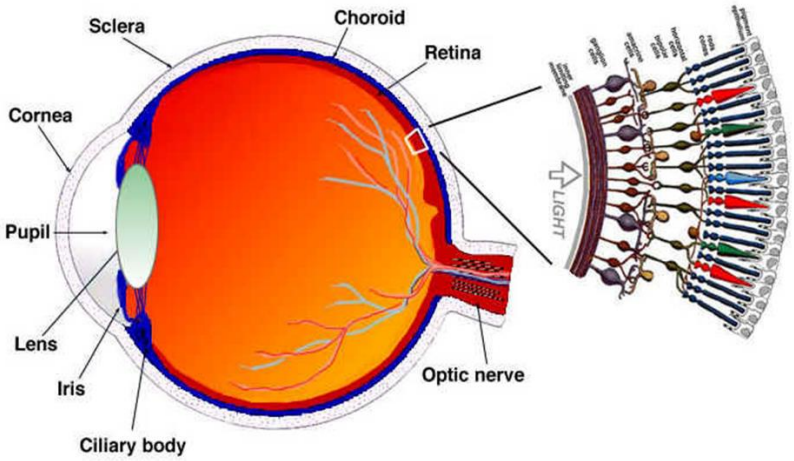
	Thermal (high or low temperatures). Internal nociceptors- found in a variety of organs, such as the muscles, the joints, the bladder, the gut, and the digestive tract. Severe heat, cold, irritating chemicals, and strong mechanical stimuli may elicit a response from nociceptors that the brain interprets as pain or itching. 
80	**Skin sensors of heat and cold**: Sensors of temperature or Thermoreceptors are also bare nerve endings. They are located in epidermis or dermis of the mammalian skin. Mammals have different areas sensitive to heat and cold. Cold receptors respond to temperature below skin temperature. Heat receptors respond to temperature above skin temperature. The ability to detect changes in temperature has become well developed in a number of animals. For example Pit organs in rattlesnakes and vipers. Heat receptors respond to temperature above skin temperature. The ability to detect changes in temperature has become well developed in a number of animals. For example Pit organs in rattlesnakes and vipers.
81	**Skin Sensors of Mechanical Stimuli** To obtain information from the environment, the animals rely on tactile stimuli. Following are the mechanical receptors found in the skin.

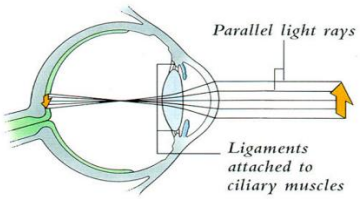
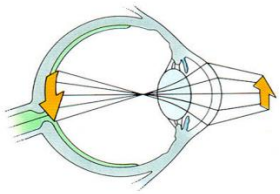
	i) Bare sensory nerve endings: These are widely distributed receptors in the vertebrate body and are involved in pain, heat and feeble pressure. ii) Tactile (Meissner's) corpuscles: They are a type of nerve endings in the skin that are responsible for sensitivity to light touch. iii) Bulbs of Krause: the recepters found in dermis in certain parts of the body that respond to physical position changes. iv) Pacinian corpuscles: These are responsible for sensitivity to vibration and pressure. v) Organs of Ruffini: Detect stress deformation within joints and warmth. Many mammals have specially adapted sensory hairs called Vibrissae on their wrists, snout, eye brows and whiskers. Around the base of each vibrissa is a blood sinus. Nerves bordering the sinus carry impulses to the brain for interpretation.  ■ **Merkel's disk** ○ Skin Curvature ■ **Meissner Corpuscle** ○ Light Touch ■ **Free Nerve Endings** ○ Pain ○ Chemicals ■ **Ruffini's ending** ○ Heat ○ Stretch ■ **Pacinian Corpuscle** ○ Deep Pressure ○ Vibrations
82	**Sonar or Echolocation** Sonar/biosonar/ Echolocation is a physiological process for locating distant objects by means of sound waves reflected back to the emitter (such as a bat) by the objects. Echolocation is used for orientation, obstacle avoidance, food procurement, and social interactions.

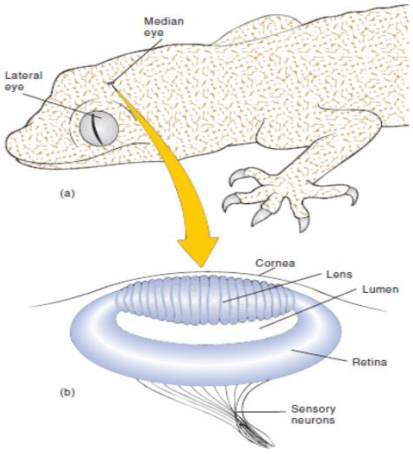
	Other animals which have this sense of sonar are, shrews, several cave dwelling birds, whales, and dolphins. These animals emit high frequency sounds which return after bouncing off objects in the environment. The returning waves from the object (e.g.flying insect) provide enough information for the bat to locate and catch the prey. This process lasts for 2-3 miliseconds and are repeated several hundred times per second
83	**Smell or olfaction:** The sense of smell or olfaction is due to the olfactory neurons (receptor cells) present in the roof of nasal cavity of the vertebrates. These receptor cells are densely packed. In dog, there are more than 40 million olfactory receptor cells per square centimeter. Each olfactory cell ends in a tuft of cilia containing receptor site for various chemicals. How odor is perceived? Odor molecules physically interact with protein receptors on the plasma membrane. This interaction alters membrane permeability and leads to generator potential.
84	**Olfactory sense in various vertebrates:** <u>In Fishes</u> Most of the fishes such as Lamprey and Salmon return to spawn in the same stream in which they hatched years earlier. Their migration in the steams often involve distances of hundreds of km. This migration is guided by fishes' perception of characteristic odor of their spawning stream <u>In amphibians</u> olfaction is used to detect noxious chemicals, food and mate. <u>In reptiles</u>

	olfaction is better adapted than in amphibians. Jacobson's (vomeronasal) organ: Apart from olfactory epithelium most reptiles possess Jacobson's organs. These organs are best developed in snakes and lizards. Jacobson's organ is a spherical structure having a narrow duct that opens into mouth. The protrusible, forked tongue of snakes is used to take sample of air borne chemicals. Snake flicks the tongue out take the sample and then moves to the Jacobson's organ which perceive odor molecules. Turtles and Tuatara use the Jacobson's organ to taste objects held in oral cavity. In birds olfactory epithelium is poorly developed, so olfaction plays minor role in the lives of birds. Exceptions are vultures. They locate dead and dying prey largely by smell. In birds olfactory epithelium is poorly developed, so olfaction plays minor role in the lives of birds. Exceptions are vultures. They locate dead and dying prey largely by smell.
85	**Taste or gustation:** Taste receptors are the chemoreceptors. Tongue is the primary organ of taste. The tongue is equipped with many taste buds housed in papillae on its dorsal surface. Papillae give the tongue a bumpy appearance. All sugars are sweet because they contain OH groups with a particular orientation that can interact with the taste receptor for sweetness in our tongues.

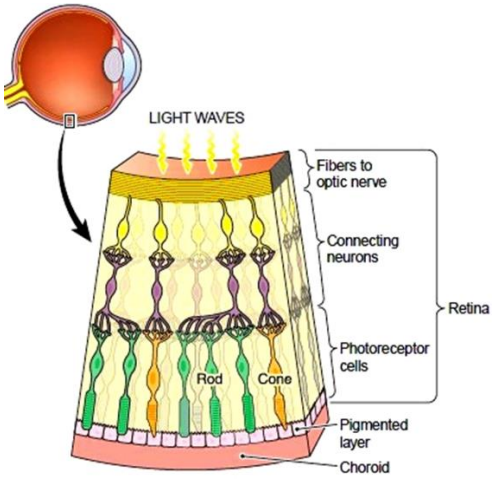
	
86	**Taste in various vertebrates** Vertebrates other than mammals may have taste buds on other parts of the body. Taste buds are inside the fish's mouth, on its tongue and scattered all over the body including barbles and fins. Vertebrates other than mammals may have taste buds on other parts of the body. Taste buds are inside the fish's mouth, on its tongue and scattered all over the body including barbles and fins. Vertebrates other than mammals may have taste buds on other parts of the body. Taste buds are inside the fish's mouth, on its tongue and scattered all over the body including barbles and fins. However, some birds have an acute sense of taste. Extensive research into bird senses has shown that vultures, seabirds, kiwis and parrots have well-developed olfactory glands, giving them some sense of smell and taste.
87	**Photoreception:** Photoreception/Vision is defined as any of the biological responses of animals to stimulation by light. Photoreception refers to mechanisms of light detection that lead to vision Vision depends on specialized light-sensitive cells called photoreceptors, which are located in the eye. The quality of vision provided by photoreceptors varies enormously among animals.

	Some simple eyes such as those of flatworms have few photoreceptors and are capable of determining only the approximate direction of a light source. The quality of vision provided by photoreceptors varies enormously among animals. Some simple eyes such as those of flatworms have few photoreceptors and are capable of determining only the approximate direction of a light source. 
88	**Anatomy of Human eye** The eye is composed of the anterior and the posterior segments. The anterior segment is made up of the cornea, iris and lens. The cornea is transparent and more curved, and is linked to the larger posterior segment. The posterior segment is composed of the vitreous, retina, choroid and the outer white shell called the sclera. The transparent cornea is continuous with the sclera and covers the front of the eye ball. Choroid tissue also extends the front of the eye ball to form iris, ciliary body and suspensory ligaments. The cornea is about 11.5 mm in diameter, and 1/2 mm in thickness. The iris is the circular pigmented structure which is composed of radial and circular muscles surrounding the center of the eye.

	The pupil is an aperture which appears to be black present in the center of iris The size of the pupil is adjusted by the iris muscles. A clear fluid, the aqueous humor fills the anterior chamber and jelly-like vitreous humor fills the posterior chamber. The retina is the innermost light sensitive layer composed of rods and cones. The size of the eye ball is about 24 mm in diameter. At birth it is 16-17 mm and attains maximum size by 12 years of age. However, the maximum increase is within three years. The moist mucous membrane that covers the eye ball is the conjunctiva. The fovea is a tiny pit in the retina aligned with the central axis of the lens. Fovea contains closely spaced cones (no rods) and produces the highest visual resolution anywhere on the retina.
89	**Accommodation of eye:** Accommodation is the adjustment of the optics of the eye to keep an object in focus on the retina as its distance from the eye varies. It is the process of adjusting the focal length of a lens. Accommodation is the ability of the eye lens to see both near and distant objects by adjusting its focal length. The minimum distance at which the eye can see objects clearly is called near point vision. The farthest at which the eye can see objects clearly is called far point vision.  **Distant objects** To focus on objects in the distance, the ciliary muscles relax and the lens flattens and thins. Light rays are slightly refracted (bent) by the lens.  **Nearby objects** To view objects that are nearby, the ciliary muscles contract and the lens becomes more rounded. The point at which the image of a close object becomes blurred is called the near point of vision; it occurs when the lens reaches its maximum curvature.

90	**Adaptation of Amphibians Eye** <u>Fish eyes:</u> Eyes of fishes are similar in most respects to the eyes of other vertebrates in both structure and function. However, fish eyes are without eye lids. Lens is spherical and close to cornea. Focusing requires moving the lens forward and backward. <u>Amphibian Eye:</u> In anurans and salamander eyes are close together on the front of head and have binocular vision. Some salamanders have smaller and lateral eyes and lack binocular vision. Laterally placed eyes permit such animals to see well off to their sides. Binocular vision is a type of vision in which an animal having two eyes is able to perceive a single three-dimensional image of its surroundings.
91	**Vision in Reptiles** Eye in Reptiles are similar in structure to those of amphibians. Most reptiles have a ring of bones around each eye that supports the eyeball except crocodiles and snakes. Some lizards and tuatara contain a third eye called a parietal eye, located on the top of their head.  This eye develops from optic tactum (midbrain)

	This eye contains a cornea, lens, and retina like the lateral eyes. Snakes do not possess eyelids. A scale that is actually part of the skin, a spectacle, protects the eye. Most reptiles possess upper and lower eye lids, nictitating membrane that protect and cleanse the surface of the eye. Usually, non-poisonous snakes have round pupils while poisonous snakes have elliptical pupils.
92	The structure of bird eye is similar to that of other vertebrates. Birds are highly visual animals with unique features and adaptations that allow them to fly. But they have a number of adaptations which give visual acuity superior to that of other vertebrate groups. Birds have unique double focusing mechanism Padlike structures control the curvature of the lens and ciliary muscles. Also change the curvature of the cornea. Instantaneous focusing of both allows the bird of prey to descend down rapidly to catch fish. judgment of distances. Nocturnal species have but a high density of rod cells which function well in poor light. Birds can see ultraviolet (UV) light because their lenses and other ocular media transmit UV. They possess a class of photoreceptor, which is maximally sensitive to violet or UV light, depending on the species.

93	**Functioning of eye** The retina is a thin layer of tissue that lines the back of the eye on the inside. The retina is a light-sensitive layer which is composed of pigmented epithelium, that covers the choroid layer. Nervous tissue that contains photoreceptor cells lies on this basement layer. The photoreceptor cells are called rod and cone cells. Rods are sensitive to dim light whereas cones respond to high intensity light and involved in color perception. With the help of the cornea and crystalline lens, image is formed on the retina which transforms it into nerve impulses and sent to the brain. Chemistry of Vision: Visual perception in humans occurs through the absorption of electromagnetic radiations by photoreceptors in the retina. When rhodopsin, a pigment in rod cells, absorb light energy. 
94	**Light, Eye Vision** Light is electromagnetic radiation that has properties of waves. The electromagnetic spectrum can be divided into several bands based on the wavelength. As we have discussed before, visible light represents a narrow group of wavelengths between about 380 nm and 730 nm.

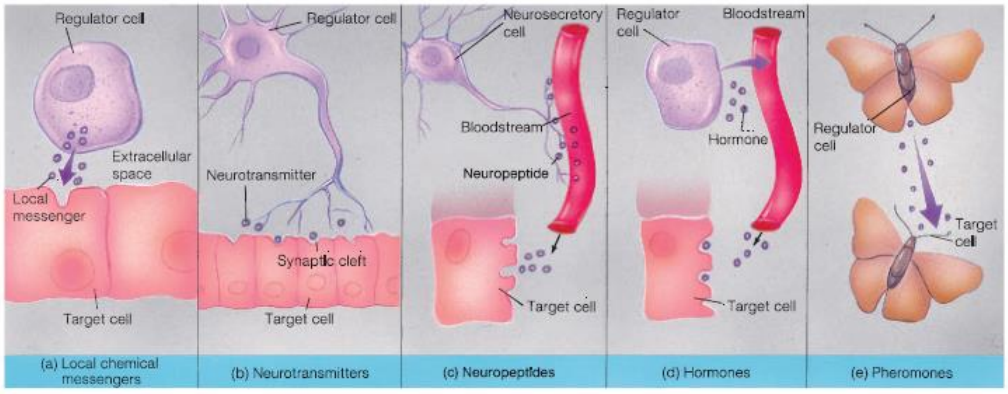
	Nature of light is said to dual i.e. it shows characteristics of both waves and particles. These particles are called photons when comes into contact with matter. Photon as the particle of light has no charge and always move at the speed of light. The photon has only energy and no mass. When a pigment (rhodopsin) in a rod cell absorbs light energy, the energy that this reaction releases triggers the generator potential in an axon and then an action potential that leaves the eyeball via the optic nerve to the brain. When the photoreceptor cells are not being stimulated (i.e., in the dark), vitamin A and energy from ATP convert rhodopsin back to its light-sensitive form. <u>Fovea</u> In the middle of the retina is a small dimple called the fovea or fovea centralis. This is the center of the eye sharpest vision and the location of most color perception. Our perception of color is based on perception of the light wavelength. Blue, yellow and red are the primary colors. These colors contain only one wavelength so they are called pure colors.
95	**Common eye defects** *Myopia (nearsightedness),* *Hyperopia (farsightedness) and* *Astigmatism are common eye defects. They are often called "refractive errors".* *Cataract* *Glaucoma* <u>Nearsightedness or Myopia,</u> is a vision condition in which close objects are seen clearly, but objects farther away appear blurred. <u>Farsightedness or hyperopia</u>, is a vision condition in which distant objects are seen clearly, but close objects does not come into proper focus. Hyperopia occurs if the eyeball is too short. Light entering the eye is therefore, not focused correctly. <u>Astigmatism</u> is a vision condition that causes blurred vision due to the irregular shape of the cornea or the curvature of the lens inside the eye. the light entering the eye is not focused correctly

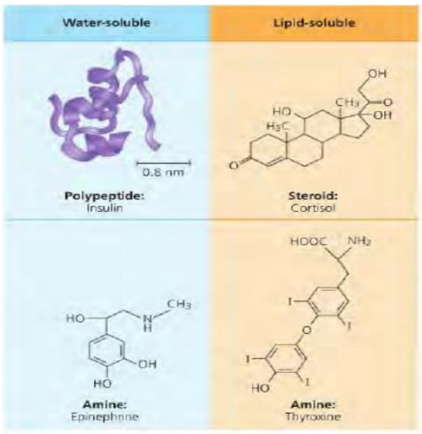
	Cataract is a condition that occurs when the inner lens of the eye becomes darkened or opaque. Glaucoma is a disease which damage the eyes optic nerve and can result in vision loss and blindness.
96	**Shining eyes:** Eye shine can be seen in many animals. Eye shine occurs in a wide variety of colors including white, blue, green, yellow, pink and red. Eye shine is a type of iridescence. E.g Soap bubbles, sea shells: Eye shine is a visible effect of the tapetum lucidum Tapetum is a layer of tissue in the eye of many vertebrates. It lies immediately behind the retina. When light shines into the eye of an animal having a tapetum, the pupil appears to glow. The tapetum lucidum functions as a retro reflector. It reflects visible light back through the retina, increasing the light available to the photoreceptors.
97	Summary of the Chapter Part I
98	Summary of the Chapter Part II
99	Summary of the Chapter Part III

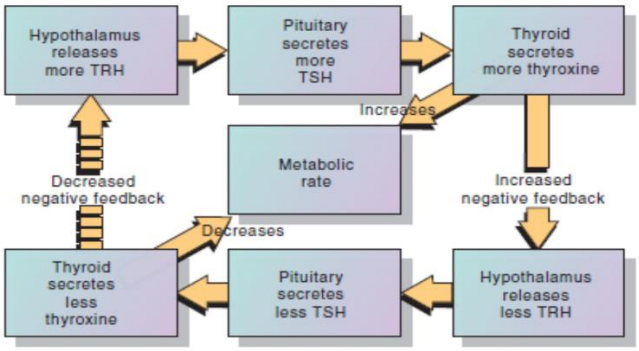
CHAPTER 25 COMMUNICATIO II **ENDOCRINE SYSTEM AND CHEMICAL MESSENGERS**	
100	**Introduction of Chemical Messengers** Chemical messengers are involved in: 1. Communication 2. Maintenance of homeostasis in an animal's body, 3. Body's response to various stimuli. One type of chemical messenger is a hormone.

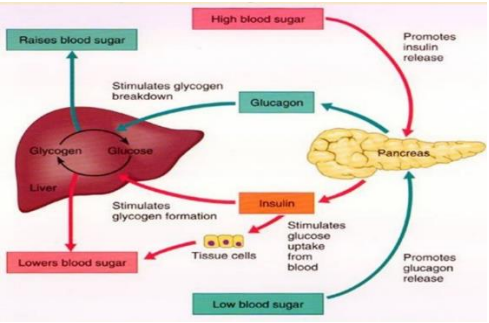
	Only those cells that have specific receptors for a hormone can respond to that hormone. Hormones work with nerves to communicate, coordinate, and integrate activities within the body of an animal. Almost every invertebrate produces hormones However, the physiology of invertebrate hormones is often quite different from that of vertebrate hormones. <u>The major endocrine glands of vertebrates include:</u> hypothalamus, pituitary, thyroid, parathyroids, adrenals, pineal, thymus. tissues, however, such as
101	**Evolution of Coordination** • Evolutionarily, new messengers are uncommon. Instead, "old" messengers are adapted to new purposes. For example, some ancient protein hormones are in species ranging from bacteria to human. Scientists suggest that chemical messengers may initially have evolved in single-celled organisms to coordinate feeding or reproduction. • As multicellularity evolved, more complex organs also evolved to govern the many individual coordination tasks, but control centers relied on the same kinds of messengers that were present in the simpler organisms. • Some of the messengers worked fairly slowly but had long-lasting effects on distant cells; these became the modern hormones. • Others worked more quickly but influenced only adjacent cells for short periods; these became the neurotransmitters and local chemical messengers. • Clearly, chemical messengers have an ancient origin and must have been conserved for hundreds of millions of years. • It seems likely that the chemical messengers. regulating growth and reproduction were among the first to appear.These messengers were probably

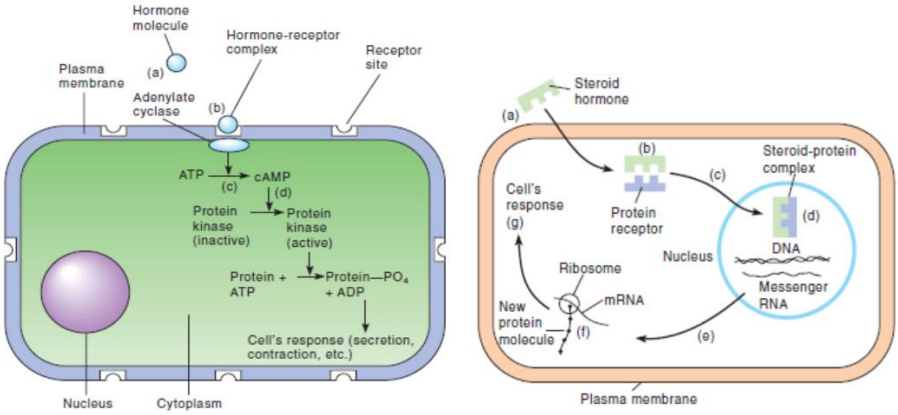
	• Later, specific hormones developed to play important regulatory roles in molting, growth, reproduction and metamorphosis, in various invertebrates. • Chemical messengers and their associated secretory structures became even more complex with the appearance of vertebrates.
102	**Chemical Messengers Part I** • The development of most animals commences with fertilization and the subsequent division of the zygote. • Further development then depends on continued cell proliferation, differentiation and growth. • Integration communication, and coordination of all physiological processes, such as metabolism, respiration, excretion, movement, and reproduction, depend on chemical messengers. • Integration communication, and coordination of all physiological processes, such as metabolism, respiration, excretion, movement, and reproduction, depend on chemical messengers.
103	**Chemical Messengers Part II** Chemical messengers can be categorized as follows: <u>i) Local chemical messengers.</u> Many cells secrete chemicals that alter physiological conditions in the immediate vicinity. Most of these chemicals act on adjacent cells and do not accumulate in the blood. Vertebrate examples include some of the chemicals called lumones that the gut produces and that help regulate digestion. In a wound, mast cells secrete a substance called histamine that participates in the inflammatory response. <u>ii) Neurotransmitters.</u> Neurons secrete chemicals called neurotransmitters (e.g., nitric oxide and acetylcholine) that act on immediately adjacent target cells. These chemical messengers reach high concentrations in the synaptic cleft, act quickly, and are actively degraded and recycled.

	iii) Neuropeptides. Some specialized neurons (called neurosecretory cells) secrete neuropeptides (neurohormones). The blood or other body fluids transport neuropeptides to nonadjacent target cells, where neuropeptides exert their effects. In mammals, for example, certain nerve cells in the hypothalamus release a neuropeptide that causes the pituitary gland to release the hormone oxytocin, which induces powerful uterine contractions during the delivery of offspring. iv) Hormones. Endocrine glands or cells secrete hormones that the bloodstream transports to nonadjacent target cells. Many examples are given in the rest of this chapter. Thyroxin, Insulin, Glucagon, FSH, LH etc. v) Pheromones. Pheromones are chemical messengers released to the exterior of one animal that affect the behavior of another individual of the same species. 
104	**Biochemistry of hormones I** Biochemically the hormones may be the: i) Proteins (polypeptides) e.g. insulin ii) Derivatives of amino acids (amines), e.g. TSH iii) Steroids e.g. Sex hormones and adrenocortical hormones. iv) A few are fatty acid derivatives. For example- Postglndins • They are synthesized in the walls of blood vessels.

	• Regulate the contraction of smooth muscle tissue. • Involved in inflammation. 
105	**Biochemistry of hormones II** General characteristics of hormones are as follows: i) Hormones are secreted by specialized cells, the endocrine cells. ii) Hormones are chemical messengers. They are chemical signals that circulate in the body fluids. iii) Hormones are directly secreted into the blood stream and are carried by the blood throughout the entire body, yet they affect only certain cells. iv) Specific cells that respond to a given hormone have receptor sites for that hormone. v) Hormones are required in very minute quantities. vi) Steroid hormones help control: Metabolism, Inflammation, Immune functions, Salt and water balance, Development of sexual characteristics and ability to withstand illness and injury.
106	**Feedback control system of hormone secretion I**

	• Feedback system is defined as the system in which the controlling mechanism is itself controlled by the product of the reaction it is controlling. e.g. incubator, toaster, iron. • The endocrine system is controlled by complex feedback mechanisms in which the secretion of a hormone is turned on (stimulated) or turned off (inhibited), depending on its concentration. • It may be Positive or Negative. • The endocrine system is controlled by complex feedback mechanisms in which the secretion of a hormone is turned on (stimulated) or turned off (inhibited), depending on its concentration. • It may be Positive or Negative. • Negative Feed back: E.g • The secretion of thyroxin by thyroid is regulated by a negative feedback loop. • When the level of thyroxin is high enough, the hormone feedback to stop the hypothalamus from secreting TRH (thyrotorpin-releasing hormone) and the pituitary from secreting TSH. 
107	**Feedback control system of hormone secretion II** • Hormones are not released in a constant, steady stream. • Rather they are secreted in fluctuations depending on the physiological demand for their actions. • A negative feedback system is used to regulate the secretion of hormones.

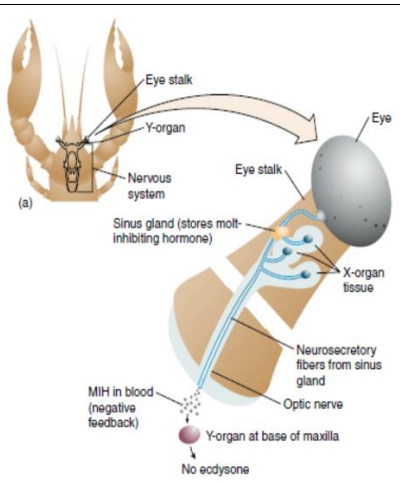
	• Such as that the release of a specific hormone is increased or decreased in response to physiologic changes. • Some hormones are released in short bursts in periods, showing daily or even monthly cycles (such as a menstrual cycle). • The regulation of blood glucose concentration is an example of how the endocrine system uses negative feedback to maintain homeostasis. • When blood glucose concentration is high, insulin is released by the pancreas in response to the negative feedback of elevated blood glucose levels • In response to insulin, cells take up glucose, thereby returning blood glucose concentration back to normal. • Insulin release is then inhibited until blood glucose levels rise again. 
108	**Mechanism of hormone action I** • Hormones modify the biochemical activity of a target cell or tissue. Two basic mechanisms are involved. <u>1. Fixed membrane-receptor mechanism.</u> <u>2. Mobile-receptor mechanism.</u> <u>*1. Fixed membrane-receptor mechanism.*</u> • This complex activates the enzyme-adenylate cyclase in the membrane. • The activated enzyme converts ATP into a nucleotide called cyclic AMP.

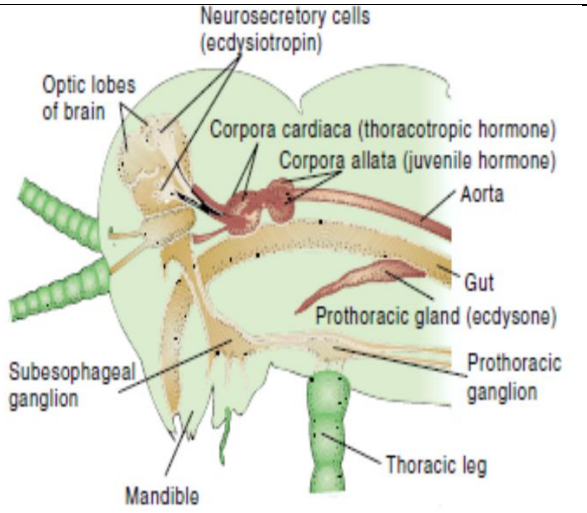
	• Cyclic AMP diffuses throughout the cytoplasm and activates an enzyme called protein kinase. • Which then phosphorylates specific proteins in the cell, thereby triggering the biochemical reaction, leading ultimately to the cell's response. • After inducing the target cell to perform its specific function, the phosphodiesterase inactivates cyclic AMP • In the meantime, the receptor on the plasma membrane loses the first messenger and now becomes available for a new reaction. 
109	**Mechanism of hormone action II** • Hormones modify the biochemical activity of a target cell or tissue. Two basic mechanisms are involved. 1. Fixed membrane-receptor mechanism. 2. Mobile-receptor mechanism. • *Mobile-receptor mechanism.* • A steroid hormone molecule (e.g., testosterone) diffuses from the blood to a target cell across the plasma membrane of the target cell. • Once in the cytoplasm, the hormone binds to a receptor that carries it into the nucleus. • This steroid-protein complex triggers transcription of specific gene regions of DNA. • The messenger RNA transcript is then translated into a gene product via protein synthesis in the cytoplasm.

	• The new protein then mediates the cell's response.
110	**Hormones of Porifera and Cnidarians** • The survival of any group of animals depends on growth, maturation, and reproduction. • Thus, chemicals regulating growth, maturation, and reproduction probably were among the first hormones to appear during the course of animal evolution. • The first hormones were probably neurosecretions. • Most of the chemicals functioning as hormones in invertebrate animals are neurosecretions called neuropeptides. • Molluscs, arthropods, echinoderms have hormones other than neurosecretions. • The **porifera** (sponges) do not have classical endocrine glands. • Since sponges do not have neurons, they also do not have neurosecretory cells. • The nerve cells of **Cnidaria** (Hydra)contain a growth-promoting hormone that stimulates budding, regeneration, and growth. For example, when the hormone is present in the medium in which • For example, when the hormone is present in the medium in which fragments of Hydra are incubated, "head" regeneration is accelerated. • This so-called "head activator" also stimulates mitosis in Hydra.
111	**Hormones of Platyhelminthes and Nemertean** **Platyhelminths:** • Zoologists identified neurosecretory cells in various flatworms These cells are in the cerebral ganglion and along major nerve cords. • The neuropeptides that the cells produce function in regeneration, asexual reproduction, and gonad maturation.

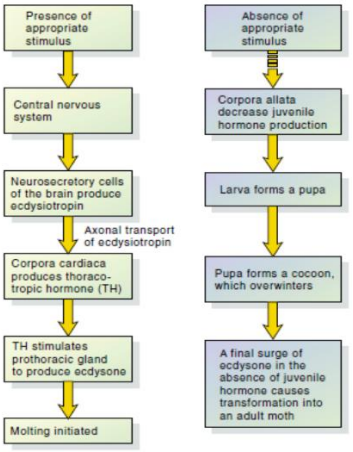
	• For example, neurosecretory cells in the scolex of some tapeworms control shedding of the proglottids or the initiation of strobilization. • NEMERTEANS Nemerteans (ribbon worms) have more cephalization than platyhelminths and a larger brain, composed of a dorsal and ventral pair of ganglia connected by a nerve ring. • The neuropeptide that these ganglia produce appears to control gonadal development and to regulate water balance.
112	**Hormones of Nematodes and Annelida:** **Nematodes:** • No classical endocrine glands have been identified in nematodes. • However, they do have neurosecretory cells associated with the central nervous system. • The neuropeptide that this nervous tissue produces apparently controls ecdysis of the old cuticle. • The neuropeptide is released after a new cuticle is produced . • It stimulates the excretory gland to secrete an enzyme (leucine aminopeptidase) into the space between the old and new cuticle. • The accumulation of fluid in this space causes the old cuticle to split and be shed. **Annelids:** • Annelids have a well-developed and cephalized nervous system, a well-developed circulatory system, and a large coelom. • Correspondingly, they have well-developed endocrine control of physiological functions. • The various endocrine systems of annelids are generally involved with morphogenesis, development, growth, regeneration, and gonadal maturation. • For example, in polychaetes, juvenile hormone inhibits the gonads and stimulates growth and regeneration. • Another hormone, gonadotropin, stimulates the development of eggs. • In leeches, a neuropeptide stimulates gamete development and triggers color changes.

	• Hormones for osmoregulation have been reported in oligochaetes. • Hormone that maintains a high concentration of blood glucose has been reported in oligochaete, Lumbricus. “Hyperglycemic hormones”
113	**Hormones of Molluscs** • The ring of ganglia that constitutes the central nervous system of molluscs is richly endowed with neurosecretory cells. • The neuropeptides that these cells produce help regulate heart rate, kidney function, and energy metabolism. • In certain gastropods, such as the common land snail Helix, a specific hormone stimulates spermatogenesis. • Another hormone, termed egg-laying hormone, stimulates egg development; and hormones from the ovary and testis stimulate accessory sex organs. • In all snails, a growth hormone controls shell growth. • In cephalopods, such as the octopus, and squid, the optic gland in the eye stalk produces one or more hormones. • These hormones stimulate gametetogenesis and the development of secondary sexual characteristics.
114	• **Hormones of Arthropoda I** • Crustacean: • The endocrine systems of advanced invertebrates (crustaceans and insects) are excellent examples of how hormones regulate growth, maturation, and reproduction. • The endocrine system of crayfish, controls functions such as ecdysis, color changes & sex determination.

	 • Here we discuss only Ecdysis. • X-organs are neurosecretory tissues in the crayfish eye stalks. • Associated with each X-organ is a sinus gland that accumulates and releases the secretions of the X-organ. • Other glands, called Y-organs, are at the base of the maxillae. • X-organs and Y-organs control ecdysis as follows. • In the absence of an appropriate stimulus, the X-organ produces molt-inhibiting hormone (MIH), and the sinus gland releases it. • The target of this hormone is the Y-organ. • When MIH is present in high concentrations, the Y-organ is inactive. • Under appropriate internal and external stimuli, MIH release is prevented, and the Y-organ releases the hormone ecdysone, which leads to molting.
115	**Hormones of Arthropods II** **Insects:** • The sequence of events in insects is similar to that of crustaceans, but it does not involve a molt-inhibiting hormone. • The presence of an appropriate stimulus to the central nervous system activates certain neurosecretory cells (pars intercerebralis) in the optic lobes of the brain.



- These cells secrete the hormone ecdysiotropin, which axons transport to the corpora cardiaca (a mass of neurons associated with the brain).
- The Corpora Cardiaca produces thoracotropic hormone, which is carried to the prothoracic glands, stimulating them to produce and release ecdysone, which induces molting, the reabsorption of some of the old cuticle and the development of a new cuticle.

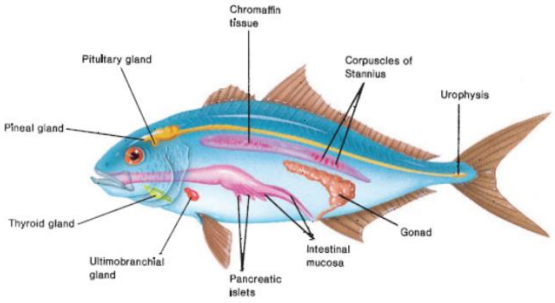


116 **Some More Hormones of Arthropods.**

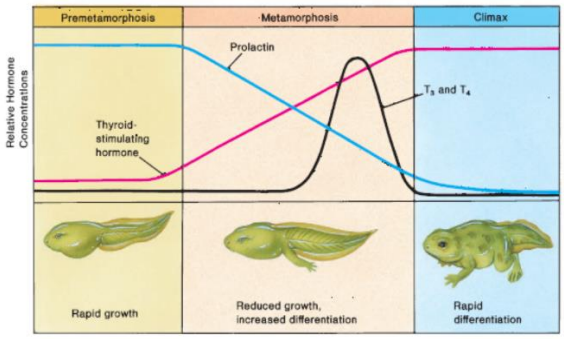
- Other neurosecretory cells in the brain and nerve cords produce the hormone bursicon
- Bursicon influences certain aspects of epidermal development, such as tanning (i.e., hardening and darkening of the chitinous outer cuticle layer).

	• Tanning is completed several hours after each molt. • Another hormone, juvenile hormone (JH), is also involved in the morphological differentiation that occurs during the molting of insects. • Just behind the insect brain are the paired corpora allata. • These structures produce JH. High concentrations of JH in the blood of an insect inhibit differentiation. • In the absence of an appropriate environmental stimulus, the corpora allata decrease JH production, which causes the insect larva to differentiate into a pupa. • The pupa then forms a cocoon to overwinter. In the spring, a final surge of ecdysone, in the absence of JH, transforms the pupa into an adult moth.
117	**Hormones of Echinodermata:** • The echinoderms are deuterostomes, they are more closely allied with chordates than are the protostome invertebrates. • However, the endocrine systems of echinoderms provide few insights into the evolution of chordate endocrine systems, because echinoderm hormones and endocrine glands are very different from those of chordates. • Zoologists do know, however, that the radial nerves of sea stars contain a neuropeptide called gonad-stimulating substance. • When this neuropeptide is injected into a mature sea star, it induces immediate shedding of the gametes, spawning behavior, and meiosis in the oocytes. • The neuropeptide also causes the release of a hormone called maturation-inducing substance, which has various effects on the reproductive system.
118	**An Overview of the Vertebrate Endocrine System** • Vertebrates have the best understood system of hormonal control. • As the earliest vertebrates evolved, hormone-producing cells and tissues developed. • Sets of nerve cells in the brain direct some endocrine tissues, such as the medullary areas of the adrenal glands.

	• Vertebrates have the best understood system of hormonal control. • As the earliest vertebrates evolved, hormone-producing cells and tissues developed. • Sets of nerve cells in the brain direct some endocrine tissues, such as the medullary areas of the adrenal glands. • Vertebrates possess two types of glands: 1. Exocrine glands, 2. Endocrine glands • Exocrine Glands: These glands secrete chemicals into ducts that, in turn, empty into body cavities or onto body surfaces (e.g., mammary, salivary, and sweat glands).
119	**Endocrine System of Fishes** • Vertebrates other than birds or mammals have somewhat similar endocrine systems, but differences do exist. • In fishes, the brain and spinal cord are the most important producers of hormones, with other glands being rudimentary. • In jawed fishes, three major regions that secrete neuropeptides: 1. Pineal gland 2. Preoptic nuclei 3. Urophysis 1. The pineal gland: It is located in the epithalamus of the brain. • It produces neuropeptides that affect pigmentation. • One specific hormone that the pineal gland produces is melatonin. • Melatonin has broad effects on body metabolism by synchronizing activity patterns with light intensity and day length. 2. The Preoptic Nuclei: These are present in the hypothalamus of the brain. • Produce various other neuropeptides that control different functions in fishes (e.g., growth, sleep, locomotion).

	3. Urophysis. • This is a discrete structure in the spinal cord of the tail. • It produces neuropeptides that help control water and ion balance, blood pressure, and smooth muscle contractions. 
120	• **Endocrine System of Amphibians.** • In many fishes, amphibians, and reptiles, hormones (e.g., melatonin) from the pineal gland control variations in skin color. When this hormone produced by one species is injected into another species, it can induce dramatic color changes • This type of experiment indicates that some hormones have a close chemical similarity despite the distant evolutionary relationships among the animals producing them. • Another example is the hormone prolactin produced by the pituitary gland). • Prolactin stimulates reproductive migrations in many animals (e.g., the movement of salamanders to water). • It also helps control water and salt balances, • Prolactin causes brooding behavior in some fishes. • It is also essential for certain saltwater fishes to enter freshwater during spawning.
121	**Hormonic Control of Metamorphosis** Hormonal Control of Frog

- When frog is immersed in water containing the hormone melatonin, light-color is exhibited.
- When melanocyte stimulating hormone is injected in frog dark color is exhibited.
- In vertebrates, thyroxine(T4) and triiodothyronine (T3) regulate metabolism. growth, and tissue differentiation.
- In amphibians, they play an additional role in metamorphosis.
- Specifically timed changes in the concentrations of three hormones—
- prolactin, thyroxine, and triiodothyronine—control metamorphosis in the frog.
- Low thyroxine and triiodothyronine concentrations and high prolactin concentrations in young tadpoles stimulate larval growth and prevent metamorphosis.
- As the hypothalamus and pituitary glands develop in the growing tadpole, the hypothalamus releases thyroid-stimulating hormone and prolactin-inhibiting hormone.
- Their release causes the pituitary gland to release thyroid-stimulating hormone and to cease production of prolactin.
- As the hypothalamus and pituitary glands develop in the growing tadpole, the hypothalamus releases thyroid-stimulating hormone and prolactin-inhibiting hormone.
- Their release causes the pituitary gland to release thyroid-stimulating hormone and to cease production of prolactin.



122	**Endocrine System of Reptiles** • Reptiles have slightly different ES as compared to Amphibians and Fishes. • The thyroids vary with the species and may be single, bilobed or paired. and the parathyroids are paired. • In turtles, snakes, and Sphenodon it is unpaired; in Crocodilia it is markedly bilobed. • Among lizards there is wide diversity in thyroid morphology and position. • Hyperparathyroidism are commonly diagnosed diseases in captive reptiles. • The thyroids are involved in control of ecdysis. • The pancreas is usually attached to the duodenum and may be pale pink in colour. • It contains alpha and beta cells that produce glucagon and insulin. • The thyroids are involved in control of ecdysis. • The pancreas is usually attached to the duodenum and may be pale pink in colour. • It contains alpha and beta cells that produce glucagon and insulin. • The cells are not grouped into islets and diabetes appears to be relatively rare. • It is both an endocrine and exocrine organ.
123	**Chromaffin Tissue and Adrenal Gland** • Chromaffin tissue is a group of neuroendocrine cells in the adrenal medulla that synthesizes steroid hormones. • These hormones are adrenaline (epinephrine) and noradrenaline (norepinephrine) • These hormones prepare some vertebrates for stressful emergency situations (fight and flight).

	• In jawless and cartilaginous fishes chromaffin tissue develops as clusters near the kidneys. • In teleosts, the chromaffin tissue is generally at the anterior end of the kidney (pronephric region). • In anurans, the chromaffin tissue is interspersed in a diffuse gland on the ventral surface of each kidney. • In lizards, the chromaffin tissue forms a capsule around the steroidogenic-producing tissue. • In birds, the chromaffin tissue is interspersed within an adrenal capsule. • In most mammals, the chromaffin tissue forms adrenal medulla, and the steroidogenic tissue forms the cortex.
124	**Endocrine System of Birds.** The endocrine glands in birds include: 1. pituitary, 2. thyroid, 3. parathyroids, 4. adrenals, 5. pancreas, 6. pineal, 7. hypothalamus, 8. thymus, 9. testes 10. ovaries 11. ultimobranchial and 12. bursa of Fabricius. • The hormones that most of these glands produce and their effects on target tissues are nearly the same as in mammals. • Some unique hormones and their functions in birds are: • In some birds (e.g., pigeons and doves), the pituitary gland secretes the hormone prolactin.

	• Prolactin stimulates the production of "pigeon's milk" by desquamation (sloughing off cells) in the pigeon's crop. • Prolactin also stimulates the development of the brood (incubation) patch • The brood patch helps keep the eggs at a temperature between 33 and 37° C. • In birds thyroxine regulates the normal development of feathers, molt cycle, and plays a role in the onset of migratory behavior. • These are in addition to the regular functioning of thyroxine. • In male birds, the testes produce the hormone testosterone. • Testosterone controls the secondary sexual characteristics of the male, such as bright plumage color, comb (when present), and spurs—all of which strongly influence sexual behavior. • The ultimobranchial glands are small, paired structures in the neck just below the parathyroid glands. • They secrete the hormone, calcitonin, which is involved in regulating blood calcium concentrations. • The bursa of Fabricius is a sac that lies just dorsal to the cloaca. • During the bird's embryological development it is well developed and it begins to shrink soon after hatching. • Its tissues produce secretions that are responsible for the maturation of white blood cells (B lymphocytes), which play an important role in immunological reactions.
125	**Endocrine System of Mammals** 1. The endocrine glands in mammals include: 2. pituitary, 3. thyroid, 4. parathyroids, 5. adrenals, 6. pancreas, 7. pineal,

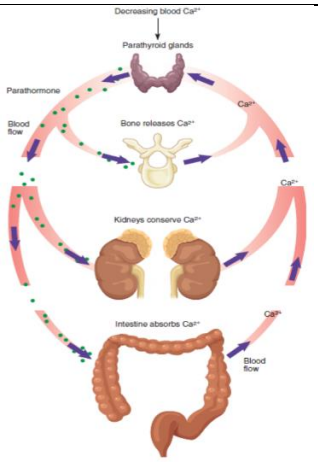
	8. hypothalamus, 9. thymus, 10. Testes 11. Ovaries • The pituitary gland (also known as the hypophysis) is directly below the hypothalamus. The pituitary has two distinct lobes: 1. the anterior lobe (adenohypophysis) and 2. the posterior lobe (neurohypophysis). • Differences Between these lobes: 1. The adenohypophysis is larger than the neurohypophysis. 2. Secretory cells called pituicytes are in the adenohypophysis, but not in the neurohypophysis. 3. The neurohypophysis has a greater supply of nerve endings.
126	• **Hormones of Adenohypophysis Part I** • The true endocrine portion of the pituitary is the adenohypophysis, which synthesizes six different hormones: • All of these hormones are polypeptides. • Produce two types of hormones: • 1. Tropic and • 2. Non tropic • **Tropic Hormones** are the hormones whose primary target is another endocrine gland. e.g TSH, ACTH, LH and FSH • **Nontropic hormones** are the growth hormone (GH) and prolactin. • (TSH) thyroid-stimulating hormone stimulates the thyroid gland to secrete thyroxine, which is the main thyroid hormone. • Adrenocorticotropic hormone (ACTH) stimulates the adrenal gland to produce and secrete steroid hormones called glucocorticoids (cortisol).

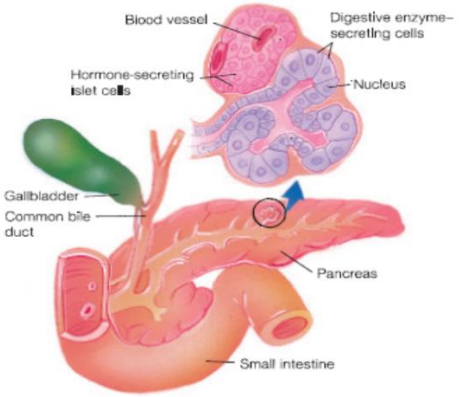
	• Secretion of ACTH is regulated by the secretion of corticotropin-releasing factor (CRF) from the hypothalamus. • CRF in turn, is regulated by a feedback system that involves factors such as stress, insulin, ADH, and other hormones. • Luteinizing hormone (LH), in females, stimulate ovulation i.e. cause the release mature ova from the ovary. • LH also stimulates corpus luteum, a temporary endocrine tissue in the ovaries. • In response the ovaries secretes the female sex hormones, the estrogen and progesterone. • In the male, the LH, stimulates Leydig cells in testes that secrete the male hormone, the testosterone. • FSH (Follicle Stimulating Hormone regulates the development, growth, pubertal maturation and reproductive processes of the body. • LH and FSH are called gonadotropins.
127	**Hormones of Adenohypophysis Part II** <u>Non Trophic Hormones are:</u> a. Growth Hormone b. Prolactin • Growth Hormone (GH) or Somatotrophic Hormone does not influence a particular target tissue. • It affects all parts of the body that are concerned with growth. • It directly induces the cell division necessary for growth. • Growth Hormone (GH) or Somatotrophic Hormone does not influence a particular target tissue. • It affects all parts of the body that are concerned with growth. • It directly induces the cell division necessary for growth. • For example, it stimulates reproductive migrations in many mammals, such as elk and caribou. • Prolactin also enhances mammary gland development and milk production in female mammals.

128	**Hormones of Neurohypophysis:** • The neurohypophysis does not manufacture any hormone. • Instead, the neurosecretory cells of the hypothalamus synthesize and secrete two hormones: 1. Antidiuretic Hormone(ADH) and 2. Oxytocin • These hormones then move down the nerve axons into the neurohypophysis, where they are stored in the axon terminals until released. • Diuretics stimulate urine excretion, whereas antidiuretics decrease urine secretion. • When a mammal becomes dehydrated, antidiuretic hormone (ADH, or vasopressin) is released and increases water absorption in the kidneys so that less urine is secreted. • Because less urine is secreted, water is retained. This negative feedback system thus restores water and solute homeostasis. • Oxytocin stimulates contraction of the uterus or uteri to aid in the expulsion of the offspring and promotes the ejection of milk from the mammary glands to provide nourishment for the newborn. • Both ADH and oxytocin are thought to have evolved from a similar ancestral chemical messenger that helped control water loss and, indirectly, solute concentrations.
129	**Pineal Gland:** • The pineal gland is also known as the third eye. • It is an endocrine gland in the Brain. • The pineal gland is located behind the third cerebral ventricle in the brain between the two cerebral hemispheres. • Darkness makes Pineal gland to produce Melatonin. • Melatonin hormone modulates wake/sleep and seasonal changes. • Light inhibits the enzymes needed for melatonin synthesis. • Because of its cyclical production, melatonin can affect many physiological processes and adjust them to diurnal and seasonal cycles.

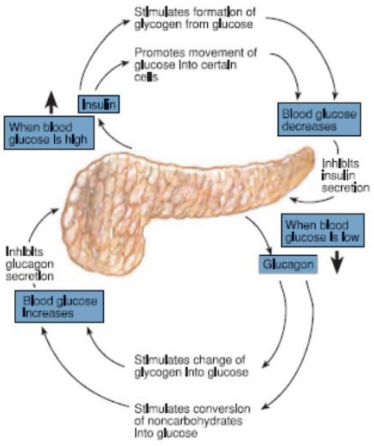
	• In humans, decreased melatonin secretion may help trigger the onset of puberty. • Melatonin is best known for the role it plays in regulating sleep patterns. • Sleep patterns are also called circadian rhythms. • If one has sleep disorder, it is the symptom that pineal gland is not producing the correct amount of melatonin. • Some health providers believe that sleep can be improved by the use of detox. • This activates the pineal gland to improve sleep and to open third eye.
130	**Thyroid gland:** • The thyroid gland is located in the neck, anterior to the trachea. • Two of its secretions are thyroxine and triiodothyronine, both of which influence overall growth, development, and metabolic rates. • Another thyroid hormone, calcitonin, helps control extracellular levels of calcium ions. • This helps promoting the deposition of these ions into bone tissue when their concentrations rise. • Once calcium returns to its homeostatic concentration, thyroid cells decrease their secretion of calcitonin. • Since the main purpose of thyroid hormone is to "run the body's metabolism," it is understandable that people with low thyroxine will have symptoms associated with a slow metabolism.
131	**Thyroid Disorders** • Several different disorders can arise when thyroid produces too much hormone ***(hyperthyroidism)*** or not enough *(hypothyroidism).* • Goitre is associated with thyroid when it is not functioning properly. • Several different disorders can arise when thyroid produces too much hormone (hyperthyroidism) or not enough (hypothyroidism). • Hyperthyroidism (overactive thyroid) occurs when thyroid gland produces too much of the hormone thyroxine.

	• Hyperthyroidism can accelerate your body's metabolism, causing unintentional weight loss and a rapid or irregular heartbeat. nervousness, anxiety, rapid hand tremor. • Goitre may occur may occure associated with a thyroid that is not functioning properly. • <u>Hypothyroidism:</u> • Its symptoms are • Fatigue. Weakness. Weight gain or difficulty losing weight (despite reduced food intake) Coarse, dry hair and dry skin.
132	**Parathyroid gland** • The parathyroid glands are tiny, pea-sized glands embedded in the thyroid lobes, usually two glands in each lobe. • The parathyroids secrete parathormone (PTH), which regulates the concentrations of calcium (Ca++) and phosphate ions in the blood. • When the Ca++ conc. in the blood bathing the parathyroid glands is low, PTH secretion increases and has the following effects: • It stimulates bone cells to break down bone tissue and release calcium ions into the blood. • It also enhances Ca++ absorption from the small intestine into the blood. • Finally, PTH promotes Ca++ reabsorption by the kidney tubules to decrease the amount of calcium excreted in the urine. • This is the example of negative feedback system for parathormone.

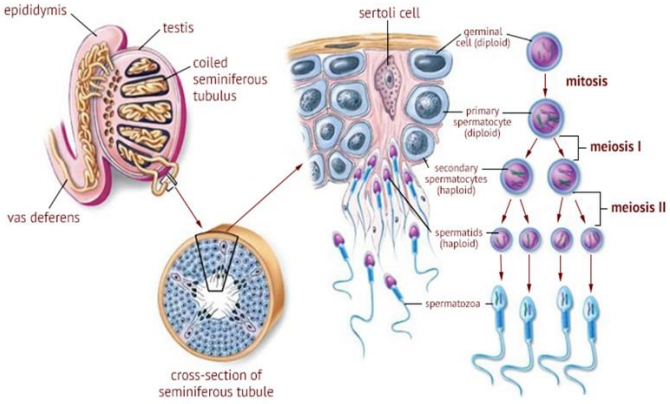
	 • Decrease in Ca++ level in blood (hypocalcemia), may spasm or twitch and cramps in arms and legs. • If you have an "acute" or sudden drop in your blood calcium level, you may notice more twitching. • Usually muscle twitching is due to stress and anxiety.
133	**Adrenal gland** • In mammals, two adrenal glands rest on top of the kidneys. • Each gland consists of two separate glandular tissues. • The inner portion, the medulla, and the outer portion the cortex. • The adrenal cortex secretes three classes of steroid hormones: 1. Glucocorticoids (cortisol), 2. Mineralocorticoids (aldosterone), and 3. Sex hormones (androgens, estrogens). • The glucocorticoids, such as cortisol, help regulate overall metabolism and the concentration of blood sugar. • They also function in defense responses to infection or tissue injury. • Aldosterone helps maintain concentrations of solutes (Na+) in the extracellular fluid. • It plays a major role in maintaining the homeostasis of extracellular fluid.

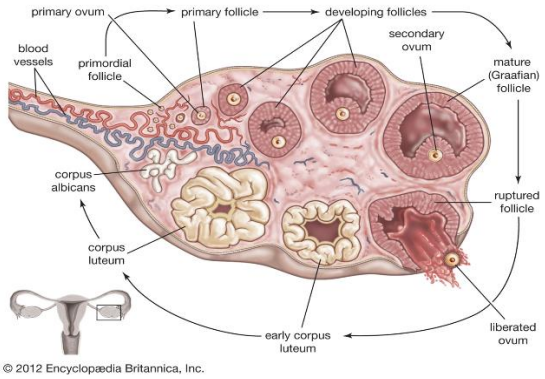
	• The sex hormones consist of male hormones called androgens(mainly dehydroepiandrosterone) • Lesser amounts of female hormones called estrogens. • The adrenal medulla is under neural control. • It contains neurosecretory cells secrete epinephrine (adrenaline) and norepinephrine (noradrenaline). • Both of which help control heart rate and carbohydrate metabolism. • In response to epinephrine and norepinephrine, the heart rate increases, blood flow increases to many vital organs. • The airways in the lungs dilate, and more oxygen is delivered to all cells of the body. • This group of events is sometimes called the "fight-or flight" response and permits the body to react strongly and quickly to emergencies.
134	**Pancreas (Glucose homeostasis)** • The pancreas is an elongated, fleshy organ posterior to the stomach. • It functions both as an exocrine and endocrine gland. • The endocrine portion of the pancreas makes up only about 1% of the gland.  • This portion synthesizes, stores, and secretes hormones from clusters of cells called pancreatic islets of Langerhans. • The pancreas contains 200,000 to 2,000,000 pancreatic islets scattered throughout the gland. Each islet contains four special groups of cells, called alpha, beta, delta and F cells.

- The alpha cells produce the hormone glucagon,
- The beta cells produce insulin.
- The delta cells secrete somatostatin, (the hypothalamic growth-hormone inhibiting factor) that also inhibits glucagon and insulin secretion.
- F-cells secrete a pancreatic polypeptide that is released into the bloodstream after a meal and inhibits somatostatin secretion, gallbladder contraction, and the secretion of pancreatic digestive enzymes.
- When glucose concentrations in the blood are high, such as after a meal, beta cells secrete insulin.
- Insulin promotes the uptake of glucose by the body's cells, including liver cells, where excess glucose can be converted to glycogen (a storage polysaccharide).
- Insulin and glucagon are crucial to the regulation blood glucose concentrations.
- When the blood glucose concentration is low, alpha cells secrete glucagon.
- Glucagon stimulates the breakdown of glycogen into glucose, which is released into the blood to raise the blood glucose level.



- The figure shows the negative feedback system that regulates the secretion of glucagon and insulin and the maintenance of appropriate blood glucose concentrations.

135	**Testes:** • In humans the testes occur as a pair of oval-shaped organs lodged in the out pocketing of the skin, the scrotum. • Testes, also called testicles, in animals, the organ that produce sperms and male hormones, the androgens.  • The gonads (ovaries and testes) secrete hormones that help regulate reproductive functions. • In the male, the testes secrete testosterone, which acts with luteinizing and follicle-stimulating hormones. • LH and FSH are produced by adenohypophysis to stimulate spermatogenesis. • Testosterone is also necessary for the growth and maintenance of the male sex organs. • Promotes the development and maintenance of sexual behavior. • In humans, it stimulates the growth of facial and pubic hair, • It also stimulates the enlargement of the larynx, which deepens the voice. • The testes also produce inhibin, which inhibits the secretion of FSH.
136	**Ovaries** • The gonads (ovaries and testes) secrete hormones that help regulate reproductive functions.

	• Four major classes of ovarian hormones help to regulate female reproductive functions 1. Estrogens (estrin, estrone, and estradiol). 2. Progestins 3. Relaxin 4. Inhibin  © 2012 Encyclopædia Britannica, Inc. • The estrogens help regulate the menstrual and estrus cycles and the development of the mammary glands and other female secondary sexual characteristics. • The progestins (primarily progesterone) also regulate the menstrual and estrus cycles, and the development of mammary glands. • It also aid in placenta formation during pregnancy. • Relaxin, which is produced in small quantities, softens the opening of the uterus (cervix) at the time of delivery. • The ovaries also produce inhibin, which inhibits the secretion of FSH.
137	**Thymus gland** • The thymus gland is a small organ behind the breastbone (sternum) that plays an important function both in the immune system and endocrine system. • It is only active until puberty. • After puberty, the thymus starts to shrink slowly and become replaced by fat. • Tymus produces, the hormone thymosin.

	• It stimulates the development of white blood cells, called T-lymphocytes, which protect the body against infections. • T-cells created by the thymus also help other organs in the immune system grow properly. • There are two types of T-cells in your body: 1. Helper T-cells and 2. Killer T-cells. • Killer T-cells do the work of destroying the infected cells. • The Helper T-cells coordinate the attack.
138	**Other Sources of Hormones Part I** 1. Placenta 2. Digestive Tract 3. Heart 4. Kidneys and 5. Adipose tissue are some more major organs that produce hormones. **<u>1. Placenta</u>** • Produces estrogens, progesterone, human chorionic gonadotropin (hCG), human chorionic somatomammotropin (hCS. • Maintain pregnancy. • Target areas are the Ovaries, mammary glands and uterus. **<u>2. Digestive Tract</u>** • Produce Secretin, Gastrin and Cholecystokinin (CCK). • Secretin stimulates the release of pancreatic juice to neutralize stomach acid. • Gastrin stimulates digestive enzymes and HCl in stomach. **<u>3. Heart</u>** • Produces the hormone, Atriopeptin that lowers blood pressure and maintains fluid balance.

	• This is in response to a rise in blood pressure, the heart releases two peptides: • A-type Natriuretic Peptide (ANP) This hormone of 28 amino acids is released from stretched atria (hence the "A"). • B-type Natriuretic Peptide (BNP) This hormone (29 amino acids) is released from the ventricles. • Streptokinase is a protein (but, despite its name, not an enzyme). • Streptokinase is used to dissolve blood clots that have formed in the blood vessels. • Because of its antigenic nature, streptokinase is quickly removed from circulation.
139	**Other Sources of Hormones Part II** **4. Kidneys** Produce three hormones: i) Erythropoietin ii) Urotensin and iii) Calcitrol i) Erythropoietin Stimulates red blood cell production, the target is bone marrow. ii) Urotensin stimulates constriction of the arteries. Vasoconstriction is needed to help maintain healthy blood flow. It keeps our body temperature from getting too cold. It can also raise blood pressure when it's necessary. iii) Calcitrol aids in the absorption of dietary calcium and phosphorus from small intestine. **5. Adipose Tissue** It produces the hormone leptin that suppresses appetite by targeting brain.
140	**Summary of the Chapter**

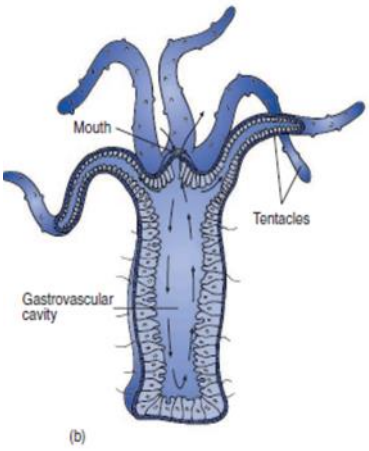
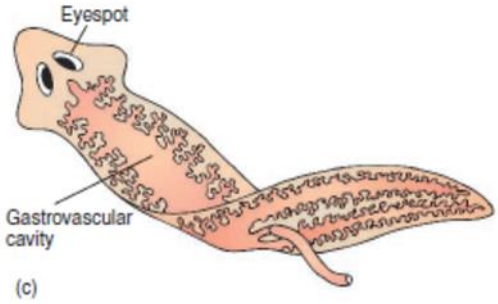
	• For smooth metabolic activity to proceed in an animal, the chemical environment of each cell must be maintained within fairly narrow limits (homeostasis). • This is accomplished using negative feedback systems. • This involves integration, communication, and coordination. • Specialized cells secrete messenger molecules. • These chemical messengers can be categorized as: Local chemical messengers (lumones), Neurotransmitters (e.g., acetylcholine), Neuropeptides, Hormones and Pheromones (e.g., sex attractant. • A hormone is a specialized chemical messenger that an endocrine gland or tissue produces. • Hormones are usually steroids, amines, proteins, or fatty acid derivatives. • Negative feedback systems often regulate hormone secretion. • Hormones modify the biochemical activity of a target cell or tissue. • Mechanisms of hormone action are: the fixed-membrane receptor mechanism (water-soluble hormones). the mobile receptor mechanism (steroid hormones). • Most of the chemicals functioning as hormones in invertebrate animals are neurosecretions called neuropeptides. • Advanced invertebrates (e.g., molluscs, arthropods, echinoderms) have non neurosecretory hormones. • In all vertebrates there is a neuroendocrine control center. • This center coordinates, communicates and integrate activities for the entire body. • This center consists of the hypothalamus and pituitary gland. • In all vertebrates there is a neuroendocrine control center.

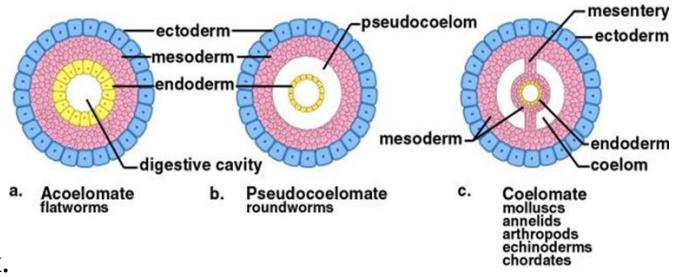
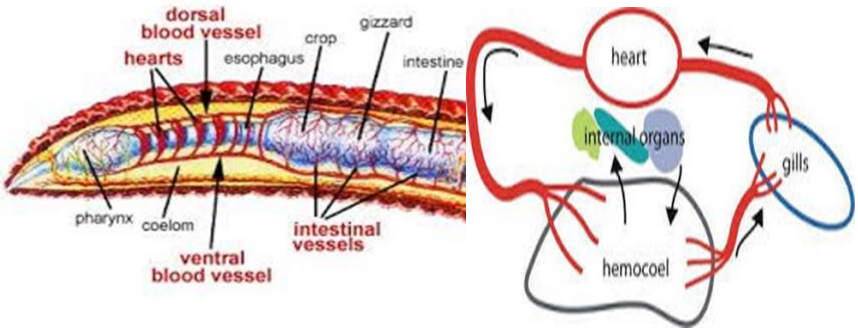
	• This center coordinates, communicates and integrate activities for the entire body. • This center consists of the hypothalamus and pituitary gland. . • In addition to these major glands, other glands and organs that carry on hormonal activity include: Placenta, Digestive tract, Heart, and Kidneys • Mind-body link occurs through the hypothalamus. • Mental states strongly affect the function of many endocrine glands. • This mind-body link occurs through the hypothalamus. • Try to describe yourself, how thoughts are transformed into physiological responses? • Compared to enzymes and genes, hormones are remarkably small molecules.

CHAPTER 26 COMMUNICATIO II

ENDOCRINE SYSTEM AND CHEMICAL MESSENGERS

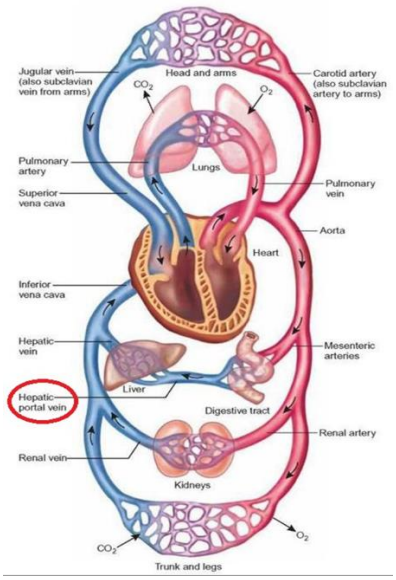
142	**Introduction of Transport in Animals (Evolutionary perspective).** **Evolutionary Perspective:** • Unicellular animals have a large surface area to volume ratio. • There is sufficient area available for the transport of substances which move into and out of the bodies. • This transport in by different ways: • Diffusion, Osmosis • Thus there is no special transport system in unicellular animals. • Same is true with simple aquatic multicellular animals. E.g. Hydra. • On the contrary, multicellular animals possess well developed and well arranged transport system in the form of blood transport system. • Thus there is no special transport system in unicellular animals. • Same is true with simple aquatic multicellular animals. E.g. Hydra. • On the contrary, multicellular animals possess well developed and well arranged transport system in the form of blood transport system.
143	**Transport Systems and its types** • Protozoa are small, with high surface area-to-volume ratios. • Plasma membrane and cytoplasm are the media through which materials such as gas, nutrients and wastes diffuse to various parts of the organism, or between the organism and the environment. • Sponges circulate water from the external environment through their bodies, instead of circulating an internal fluid. • Cnidarians, such as Hydra, have a fluid-filled internal gastro vascular cavity. • Sponges circulate water from the external environment through their bodies, instead of circulating an internal fluid. • Cnidarians, such as Hydra, have a fluid-filled internal gastro vascular cavity.

144	**Transport System in Protozoa, Porifera, Cnideria** 
145	**TS in Flatworms and Pseudocoelomates**  • Diffusion is effective for the transport of nutrients, gases, and wastes. • Body movement helps distribute materials to various parts of the body. • Pseudocoelomate invertebrates, such as rotifers, gastrotrichs, and nematodes, use the coelomic fluid of their body cavity for transport.

	• Coelomic fluids, are in direct contact with the internal organs and, produce  adequate transport. • A few other invertebrates (e.g., ectoprocts, echinoderms) also depend largely on the body cavity as a coelomic transport chamber.
146	**Transport System in Annelida, Mollusca and Arthropoda** • The annelids, such as the earthworm, have a closed circulatory system in which blood travels through vessels delivering nutrients to cells and removing wastes. • Most molluscs and arthropods have open circulatory systems.  • In Insects heart pumps hemolymph through vessels that open into a body cavity (hemocoel). • Hemolymph directly bathes the cells and tissues rather than being carried only in vessels.
147	**Coelomic Fluid, Hemolymph and Blood Cells in Invertebrates** • Some invertebrates e.g. annelids, echinoderms, use coelomic fluid as a sole circulatory or supplementary system. • Coelomic fluid may be identical in composition to interstitial fluids or may differ, particularly with respect to specific proteins and cells. • Coelomic fluid transports gases, nutrients, and waste products.

	• It also may function as a hydrostatic skeleton in certain invertebrates such as in annelids. • Hemolymph is the circulating fluid of animals with an open circulatory system. • Most arthropods, ascidians, and many molluscs have hemolymph. • In these animals, a heart pumps hemolymph at low pressures through vessels to tissue spaces (hemocoel) and sinuses. • Generally, the hemolymph volume is high and the circulation slow. • Most arthropods, ascidians, and many molluscs have hemolymph. • In these animals, a heart pumps hemolymph at low pressures through vessels to tissue spaces (hemocoel) and sinuses. • Generally, the hemolymph volume is high and the circulation slow. • In certain jumping spiders, hydrostatic pressure of the hemolymph provides a hydraulic mechanism for limb extension.
148	**Blood Cell in Invertebrates:** • The coelomic fluid, hemolymph, or blood of most animals contains circulating cells called blood cells or hemocytes. • Some cells contain a respiratory pigment, such as hemoglobin, and are called erythrocytes or red blood cells. • These cells are usually present in high numbers to facilitate oxygen transport. • Cells that do not contain respiratory pigments have other functions, such as blood clotting. • The number and types of blood cells vary dramatically in different invertebrates. • For example, annelid blood contains hemocytes that are phagocytic. • The coelomic fluid contains a variety of coelomocytes such as: a. amoebocytes, b. eleocytes, c. lampocytes and d. linocytes

	• They all function in phagocytosis, glycogen storage, encapsulation, defense responses, and excretion. • The hemolymph of molluscs has two general types of hemocytes: a. Amoebocytes and b. Granulocytes) • Apart from the functions already mentioned, they have another function called nacrezation (pearl formation) in some bivalves. • Natural Pearls form when an irritant - usually a parasite is coated by a fluid as a defense. Layer upon layer of this coating, called 'nacre', is deposited until a lustrous pearl is formed. • Insect hemolymph contains large numbers of various hemocyte types that function in phagocytosis, encapsulation, and clotting.
149	**Transport System in Vertebrates** • All vertebrates have a closed circulatory system in which the blood never leaves the blood vessels. • Blood moves from the heart, through arteries, arterioles, capillaries, venules, veins, and back to the heart. • Exchange between the blood and extracellular fluid only occurs at the capillary level. • The heart is the center to the circulatory system. • It is fist-sized pump that circulates the blood throughout the body. • As animals became more multicellular and complex the circulatory system evolved because simple diffusion was insufficient to supply all of the cells with nutrients. • The coordination of the circulatory system and the respiratory system to ensure proper gas exchange is very important in animals that have lungs and gills.



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Blood Portal Systems in Vertebrates

- Portal system is a system of blood vessels that carry blood from one bed of capillaries to the other bed of capillaries.
- Three portal systems are common:

a. Hepatic portal system

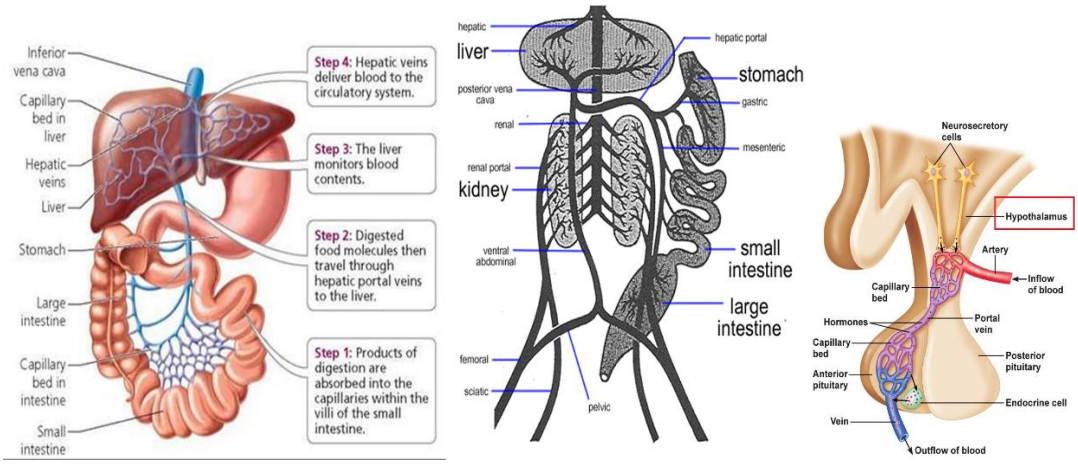
b. Renal portal system

c. Hypophysial portal System

<u>Hepatic Portal System</u>

- Nutrients from the small intestine and the right half of the colon are absorbed into blood through the bed of capillaries.
- This blood then moves to the liver through the Hepatic Portal Vein to store and metabolize in liver.
- From the liver the blood enters main blood stream via hepatic vein into the inferior vena cava to the heart.

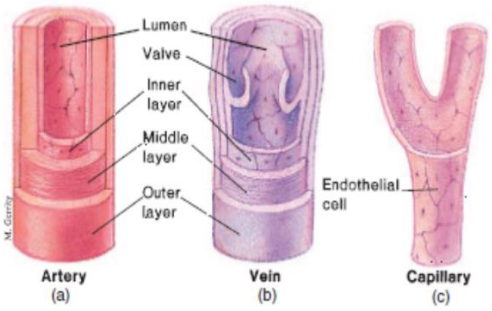
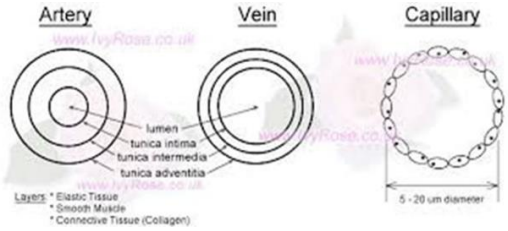
<u>Renal Portal System</u>

	• The veins which carry blood to capillary system in kidneys constitute the renal portal system. • The renal portal vein collects blood from all parts of the hind limbs and enters the kidney by number of branches from outer margin. • Inside the kidney these veins divide and re divide to form capillaries. These capillaries form 5 to 6 renal veins on emerging from the kidneys and open into post caval vein. All these veins forms renal portal system. <u>Hypophysial System:</u> • A portal system between the hypothalamus and the anterior pituitary gland. 
151	**Vertebrate Blood** • Blood (circulatory fluid) • Blood is the medium in which the dissolved nutrients, gases, hormones and wastes are transported through the body. • Blood defends against harmful microorganisms, cells and viruses. • It also prevents blood loss through coagulation (clotting); and helps regulate body temperature and pH. • It is a specialized type of connective liquid tissue which is composed of two main components: a. Fluid matrix called plasma and b. Formed elements- RBCs, WBCs and Platelets. <u>Plasma</u> is straw-colored liquid.

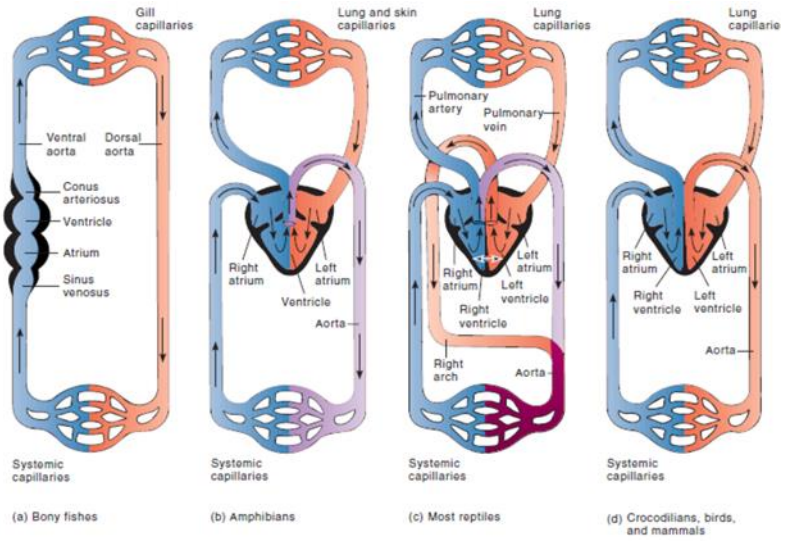
	• In mammals, plasma is composed of: a. Water (about 90%) b. A group of proteins (albumin, fibrinogen and globulins)-7%. • The remaining 3% of plasma is composed of: i) Electrolytes, amino acids, glucose and other nutrients, various enzymes, hormones, metabolic wastes, and traces of many inorganic and organic molecules. ii) Albumin represents about 60% of the total plasma proteins, it plays important roles with respect to water movement. iii) Fibrinogen is necessary for blood coagulation (clotting), and the iv) Globulins include the immunoglobulins, which functions in the immune response because it consists mostly of antibodies. • Serum is plasma from which fibrinogens have been removed.
152	• **The Vertebrate Blood Cells I (formed-element fraction consists of)** 1. Erythrocytes (red blood cells or RBCs), 2. Leukocytes (white blood cells or WBCs 3. Thrombocytes (Plate lets) **Erythrocytes or RBCs** • RBCs vary dramatically in size, shape, and number in the different vertebrates. • For example, the RBCs of most vertebrates are nucleated, but mammalian RBCs are enucleate. • Among all vertebrates, the salamander, Amphiuma has the largest RBC. • Among birds, the ostrich has the largest RBC. • Most mammalian RBCs are biconcave disks however, the camel and llama have elliptical RBCs. • 95% of mass of a RBC is hemoglobin, an iron-containing oxygen carrying protein. • The remaining 5% consists of enzymes, salts and other proteins. • The major function of hemoglobin is to transport oxygen in the form of oxyhemoglobin, from the environment to body tissues.

	• Hemoglobin also carries waste carbon dioxide (in the form of carbaminohemoglobin) from the tissues to the lungs (or gills) for removal from the body.
153	• **Vertebrate Blood Cells II** **Leucocytes White Blood Cells** are scavengers that destroy microorganisms at infection sites, remove debris that results from dead or injured cells. • All WBCs are derived from immature cells (called stem cells) in bone marrow by a process called hematopoiesis. • RBCs can be grouped into two main types: a. Granulocytes b. Agranulocytes <u>Granulocytes</u> have granules in their cytoplasm and they include: i) Neutrophils ii) Eosinophils iii) Basophils *Neutrophils* are the most numerous of the white blood cells. They are chemically attracted to sites of inflammation and are active phagocytes. *Eosinophils* are phagocytic, and ingest foreign proteins and immune complexes rather than bacteria. • Eosinophils in mammals, also release chemicals that counteract the effects of certain inflammatory chemicals released during allergic reactions. • Eosinophils are phagocytic, and ingest foreign proteins and immune complexes rather than bacteria. • Eosinophils also release chemicals that counteract the effects of certain inflammatory chemicals released during allergic reactions. *Basophils* are the least numerous WBC. • When they react with a foreign substance, their granules release histamine and heparin.

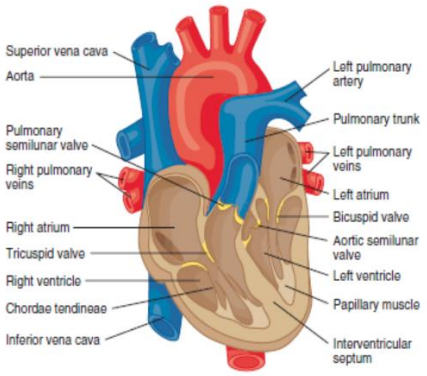
	• Histamine causes blood vessels to dilate and leak fluid at a site of inflammation, and heparin prevents blood clotting. Agranulocytes have clear cytoplasm. These include: i) Monocytes and ii) Lymphocytes. • *Monocytes* give rise to macrophages which destroy larger particles by phagocytosis. • *Lymphocytes* are of types of two types: B cells and T cells, both of which are central to the immune response. • B cells originate in the bone marrow and colonize the lymphoid tissue, where they mature. • T cells are associated with and influenced by the thymus gland before they colonize lymphoid tissue and play their role in the immune response. **Thrombocytes** or Platelets are disk-shaped cell fragments that initiate blood clotting. • When a blood vessel is injured, platelets immediately move to the site and clump, attaching themselves to the damaged area, and thereby beginning the process of blood coagulation.
154	**Vertebrate Blood Vessels** • The blood vessels are the components of the circulatory system that transports blood throughout the vertebrate body. • As blood leaves the heart it travels from arteries to arterioles to capillaries to venules to veins which return it finally to the heart. <u>Arteries</u> are elastic blood vessels that carry blood away from the heart to the organs and tissues of the body. • The central canal of an artery (and of all blood vessels) is a lumen. • Surrounding the lumen of an artery is a thick wall composed of three layers. <u>Veins</u> are relatively inelastic.

	• The wall of a vein contains the same three layers (tunicae) as arterial walls, but the middle layer is much thinner. • One or more valves are present which permit blood flow only in one direction. Capillaries are generally composed of a single layer of endothelial cells and are the most numerous blood vessels in an animal's body • An abundance of capillaries makes an enormous surface area available for the exchange of gases, fluids, nutrients, and wastes between the blood and nearby cells.  
155	**The Hearts and Circulatory Systems of Bony Fishes and Amphibians.** • The bony fish heart has two chambers—the atrium and ventricle. • Blood leaves the heart via the ventral aorta, which goes to the gills. • In the gills, blood becomes oxygenated, loses carbon dioxide, and enters the dorsal aorta. • The dorsal aorta distributes blood to all of the body organs, and then blood returns to the heart via the venous system. • Because blood only passes through the heart once, this system is called a single circulation circuit. • The dorsal aorta distributes blood to all of the body organs, and then blood returns to the heart via the venous system. • Because blood only passes through the heart once, this system is called a single circulation circuit. • The blood pumped out to the rest of the body is thus highly oxygenated.

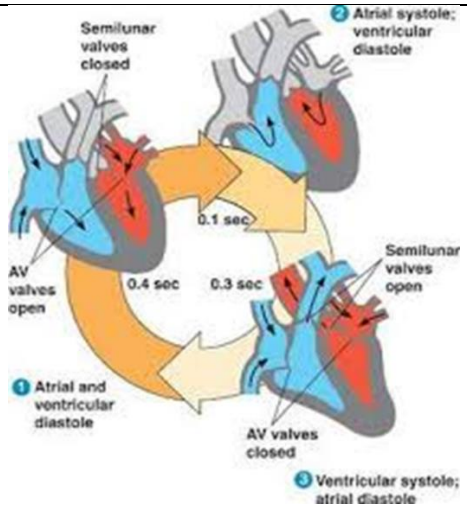
	• In the heart of most reptiles, the ventricle is partially divided into a right and left side. • Oxygenated blood from the lungs returns to the left side of the heart via the pulmonary vein and does not mix much with deoxygenated blood in the right side of the heart. • When the ventricles contract, blood is pumped out two aortae for distribution throughout the body, as well as to the lungs. • The incomplete separation of the ventricles is an important adaptation for reptiles, such as turtles. • Because it allows blood to be diverted away from the pulmonary circulation during diving and when the turtle is withdrawn into its shell. • This conserves energy and diverts blood to vital organs during the time when the lungs cannot be ventilated.
156	• **The hearts and circulatory systems of birds and mammals.** • Because it allows blood to be diverted away from the pulmonary circulation during diving and when the turtle is withdrawn into its shell. • This conserves energy and diverts blood to vital organs during the time when the lungs cannot be ventilated. • It then returns the oxygenated blood to the heart for distribution to the rest of the body. • The systemic circuit supplies all the cells, tissues, and organs of the body with oxygen-rich blood and returns oxygen-poor blood to the heart.
157	• **Comparative perspective of circulatory systems of various vertebrates.** • In bony fishes, the heart's two chambers (atrium, ventricle) pump in series. Respiratory and systemic circulations are not separate. • The amphibian heart has two atria and one ventricle. Blood from the lungs enters the left atrium, and blood from the body enters the right atrium. • The blood from both atria empties into one ventricle, which then pumps it into the respiratory and systemic circulations. • Most reptiles exhibit a greater degree of anatomical division of the ventricle into two halves. • In crocodilians, birds, and mammals, the ventricle is completely divided, forming a four-chambered heart.

	• Blood flow through the lungs completely separated from the flow to other tissues. 
158	**Vertebrate lymphatic system.** • The lymphatic system is part of the vascular system. • The system begins with network of lymphatic vessels that carry a clear fluid called lymph. • The small lymphatic capillaries merge to form larger lymphatic vessels called lymphatics. • Lymphatics are thin-walled vessels with valves that ensure the one-way flow of lymph. • Lymph is the extracellular fluid that seeps from the bloodstream and accumulates in the lymph vessels. • The lymphatic vessels pass through the lymph nodes on their way back to the heart. • Lymph nodes concentrate in several areas of the body and act as disposal centers for foreign particles and cellular debris. • They play an important role in the body's defense against disease. • The system also help transport lipids that have been absorbed from the small intestine.

	• In addition to the lymph nodes, the mammalian lymphatic system consists of lymphoid organs—the spleen the thymus gland, tonsils, and adenoids. • In birds the bursa of Fabricius is an additional lymphoid tissue. • The system also help transport lipids that have been absorbed from the small intestine. • In addition to the previously mentioned parts, the lymphatic system of birds and mammals consists of lymphoid organs—the spleen and or the thymus gland, tonsils, and adenoids in mammals.
159	**Difference b/w lymph and blood** • Lymph is colorless fluid where as Blood is colored fluid. • Lymph is the part of lymphatic system whereas blood is the part of Blood circulatory system. • Lymph helps in the body's defense as it is the part of immune system. Whereas blood is associated with circulation of O_2, CO_2. • Lymph contains plasma and a lesser number WBCs and platelets, whereas blood contains plasma, RBCs, WBCs and Platelets. • Lymph contains low amount of O_2, blood carries large ampount of O_2. • Lymph lacks protein, whereas blood plasma has Ca^{+2} & $PO4^{-3}$. • Flow of lymph is slow where flow of blood is fast. • Lymph clots slowly as less fibrinogen is present, where as blood clots quickly as high high amount of fibrinogen is present.
160	**Structure of Human Heart.** • The human heart is a hard-working pump located in the chest cavity. • Heart is enclosed in double membrane sac called pericardium. • Space between these membranes is filled up with pericardial fluid. • The wall of the heart is composed of three layers:

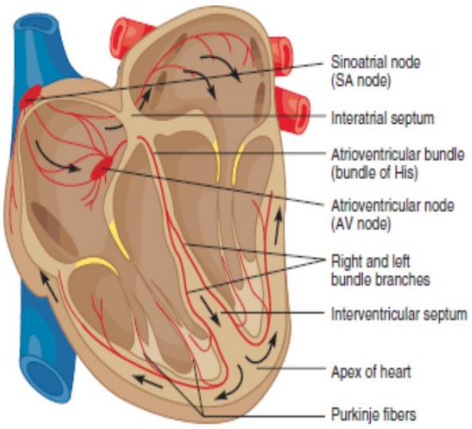
	 i) Epicardium(Connective .tissue) ii) Myocardium (cardiac muscles)and iii) Endocardium (single layer of cells). • The heart has four chambers: 2 atria and 2 ventricles) • The 2 atria are separated by Inter atrial septum. • The two ventricle are separated by Inter ventricular septum. • The two atria communicate with the ventricles through atrioventricular apertures. • On the right side the apertures is provided with Tricuspid valve. • On the left side it is bicuspid valve. • The flaps of the valves are connected to the papillary muscles by means of chordae tendenae. • The right atrium receives the inferior and superior vena cavae opening separately. • The left atrium receives pulmonary veins, 2 from each lung. • From the right ventricle arises the pulmonary artery and from the left ventricle, the Aorta. • At the point of origins of aorta and pulmonary artery there are valves called semilunar valves.
161	**Pumping Action of Human Heart.** • Adult human heart beats or pumps 72 times a minute. • In the beating of the heart the atria contract simultaneously and ventricles immediately after words.

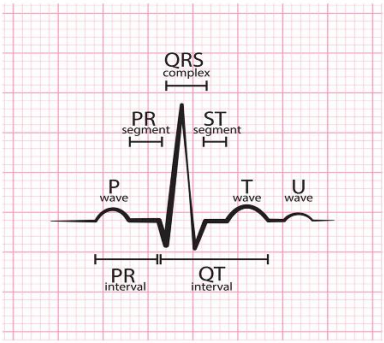
	• The deoxygenated blood from the lower and upper part of the body is brought into the right atrium. • Adult human heart beats or pumps 72 times a minute. • In the beating of the heart the atria contract simultaneously and ventricles immediately after words. • The deoxygenated blood from the lower and upper part of the body is brought into the right atrium. • Adult human heart beats or pumps 72 times a minute. • In the beating of the heart the atria contract simultaneously and ventricles immediately after words. • The deoxygenated blood from the lower and upper part of the body is brought into the right atrium. • On the contraction of the ventricles the blood from the right ventricle is taken to the lungs by the pulmonary artery. • The valves open and close due to blood pressure changes when the heart contracts during each heartbeat. • Like the valves in veins, heart valves keep blood moving in one direction, preventing backflow.
162	• **The Cardiac Cycle** • The alternate contraction and relaxation of the heart chambers is called Cardiac Cycle. • The contraction phase of the cycle is called Systole and the relaxation phase is called Diastole. • Average heartbeat of human when at rest is 72 beats a minute.



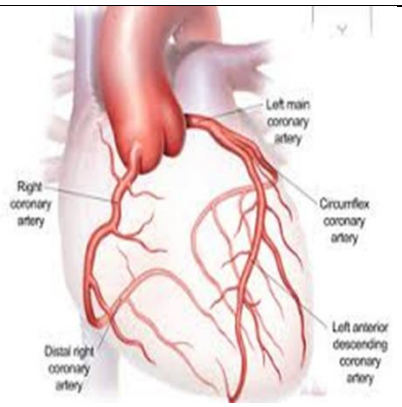
- One cardiac cycle take about 0.8 sec.
- Heart beat can be differentiated into three phases.
- Relaxation Phase: This phase is also called atrial and ventricular diastole.
- During this phase the walls of the atria and ventricle relax,
- The deoxygenated blood enters the right atrium through the vena cava and at the same time the oxygenated blood enters the Left atrium through the pulmonary vein.
- As the atria are filled up with blood, they have more pressure than the ventricles.
- Atrial Systole and Ventricular Diastole: This phase lasts 0.1 sec. During this phase atria contract simultaneously forcing all the blood out of atria into the relaxed ventricles.
- Atrial diastole and ventricular systole: This phase lasts 0.3 sec. During this phase of the cycle both the ventricles contract simultaneously pumping the blood to pulmonary arteries and aorta.
- This follows the first phase, the relaxation phase, again which lasts 0.4 sec.
- This completes the cardiac cycle extending over 0.8 sec.
- The heart of an average adult beats more than 100,000 times without fatiguing.

163	• **Electrical conduction system of the human heart**

	• Cardiac cycle is the sequence of muscle contractions and relaxations. • A "pacemaker,'' is a small mass of tissue called the sino atrial node (SA node) at the entrance to the right atrium, initiates each heartbeat. • The SA node initiates the cardiac cycle by producing an action potential that spreads over both atria, causing them to contract simultaneously. • The action potential then passes to the atrioventricular node (AV node), near the inter atrial septum.  • From here, the action potential continues through the atrio ventricular bundle (bundle of His), at the tip of the inter ventricular septum. • The atrioventricular bundle divides into right and left branches, which are continuous with the Purkinje fibers in the ventricular walls. • Stimulation of these fibers causes the ventricles to contract almost simultaneously and eject blood into the pulmonary and systemic circulations.
164	**Electrocardiogram (EKG)** • ECG is the record of electrical activity or action potential over the surface of the heart as waves called P,Q,R,S and T. currents • The electrical activity is produced when impulses travel through cardiac muscles during the heart cycle. • ECG is the record of electrical activity or action potential over the surface of the heart as waves called P,Q,R,S and T. currents

	• The electrical activity is produced when impulses travel through cardiac muscles during the heart cycle. • ECG gives important information concerning the spread of excitation to different parts of the heart. • ECG is of great value in the diagnosis of cases of abnormal cardiac rhythm and myocardial damage. • EGC is recoded by a device called elctrocariograph. • ECG gives important information concerning the spread of excitation to different parts of the heart. • ECG is of great value in the diagnosis of cases of abnormal cardiac rhythm and myocardial damage. • EGC is recoded by a device called elctrocariograph. NORMAL ECG 
165	**Blood pressure** • Blood Pressure is the force that the blood exerts against the wall of the vessels. • The pressure produced by the contraction of heart is called systolic pressure. • Pressure in the arteries when the ventricles relax is the diastolic pressure. • Blood pressure is the main force that propels blood from the heart through the arteries to arterioles to the capillary beds. • Conventionally, the BP of a healthy person is expressed as 120/80 i.e. Systolic pressure /Diastolic pressure mmHg. • Blood pressure is the main force that propels blood from the heart through the arteries to arterioles to the capillary beds.

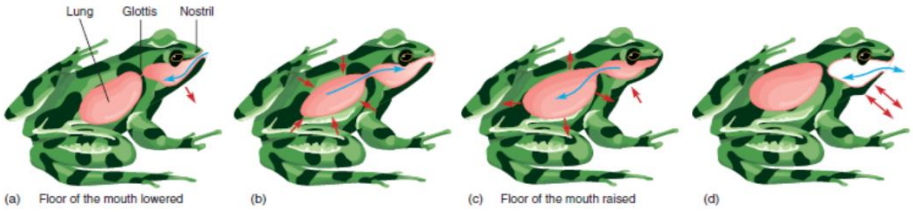
	• Conventionally, the BP of a healthy person is expressed as 120/80 i.e. Systolic pressure /Diastolic pressure mmHg. • Normal Blood Pressure Range: • 130/85 High Normal BP • 120/80 Normal BP • 110/75 Low Normal BP
166	• **Some Cardiovascular disorders** • <u>Leukemia</u> • It is the result of uncontrolled production of WBCs. • It is caused by a mutation in the cells of bone marrow called myelogenous cells. • These cell without completing the cell cycle, spread to other parts of the body by blood stream. • This process is called metastasis. • The undifferentiated cells are defective and produce at a faster that normal rate. • It is a serious disorder. The patient needs regular change of blood. • It can be cured by bone marrow transplant. • <u>Thalassemia</u> It is hereditary anemia produced by defective production of of alpha or beta Hb polypeptide chain. • Thalassemia is characterized by fatigue, severe anemia, enlargement of heart and jaundice. • It can be treated by bone marrow transplant. <u>Myocardial Infarction</u> commonly called heart attack. • Occurs when blood flow decreases or stops to a part of the heart, causing damage to the heart muscle. • The most common symptom is chest pain or discomfort which may travel into the shoulder, arm, back, neck or jaw.

	
167	• Gas Exchange: Respiratory Surfaces • Food is rich source of chemical energy. • How do animals utilize this chemical energy into useable form? • First, they must digest the organic matter so that it can enter the cells. • Second, they must provide cells with an adequate supply of oxygen. • This is required for aerobic respiration and a way of eliminating the carbon dioxide that aerobic respiration produces. • This process of gas exchange with the environment, also called external respiration or breathing is the subject of the rest of this chapter. • Respiratory Surfaces: Protists and animals have five main types of respiratory systems (surfaces): • simple diffusion across plasma membranes, • tracheae • Skin or cutaneous (integument or body surface) exchange, • gills and • lungs. • Each of these surfaces will be discussed in the next modules.
168	• **Invertebrate Respiratory Surfaces I** • Properties of Respiratory surfaces in animals: • Large surface area

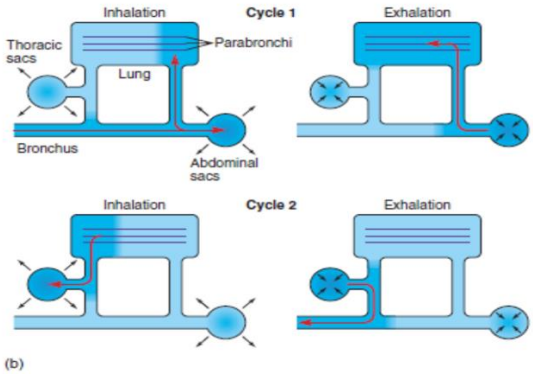
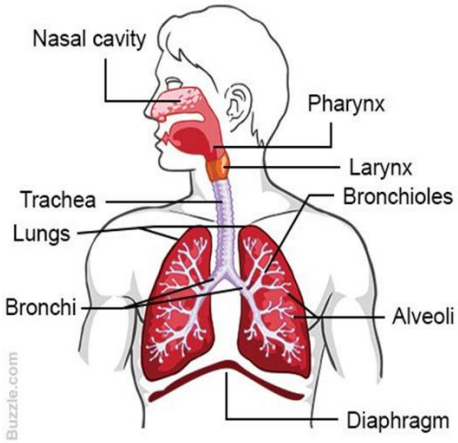
	• Moist • Thin epithelium • Ventilation • Capillary network • In protozoa, gaseous exchange take place by diffusion across the plasma membrane • Some multicellular invertebrates e.g. Hydra gases diffuse into and out of the animal cells through the entire inner and outer surfaces, which are in contact with water. • In flat worms e.g. Planarian, body is thin-walled. Again, gases diffuse into and out of the animal. • In earthworms that live in moist environments use integumentary exchange. • Earthworms have capillary networks just under their integument, and they exchange gases with the air. • In flat worms e.g. Planarian, body is thin-walled. Again, gases diffuse into and out of the animal. • In earthworms that live in moist environments use integumentary exchange. • Earthworms have capillary networks just under their integument, and they exchange gases with the air.
169	• **Invertebrate Respiratory System II** • Crustaceans and molluscs have gills that are compact and protected with hard covering devices. • Such gills divide into highly branched structures to maximize the area for gas exchange. • Crustaceans and molluscs have gills that are compact and protected with hard covering devices. • Such gills divide into highly branched structures to maximize the area for gas exchange. • Tracheae open to the outside of the body through spiracles, which usually have some kind of closure device to prevent excessive water loss.

	• Spiracles lead to branching tracheal trunks that eventually give rise to smaller branches called tracheoles, whose blind ends lie close to all cells of the body. • Tracheae open to the outside of the body through spiracles, which usually have some kind of closure device to prevent excessive water loss. • Spiracles lead to branching tracheal trunks that eventually give rise to smaller branches called tracheoles, whose blind ends lie close to all cells of the body. • Arachnids possess tracheae, book lungs, or both. • Book lungs are paired invaginations of the ventral body wall that are folded into a series of leaf-like lamellae. • The Pulmonate Lung. The mantle cavity of the pulmonate snail, Lymnaea, is highly vascularized and functions as a lung. • Arachnids possess tracheae, book lungs, or both. • Book lungs are paired invaginations of the ventral body wall that are folded into a series of leaf-like lamellae. • The Pulmonate Lung. The mantle cavity of the pulmonate snail, Lymnaea, is highly vascularized and functions as a lung.
170, 171	**Vertebrate Respiratory System** • There are two models of breathing in vertebrates: i) Bimodal breathing ii) Trimodal breathing • Bimodal breathing is the ability of an organism to exchange respiratory gases simultaneously with both air and water. e.g., some salamanders, crabs, barnacles, bivalve molluscs, and lung fishes. • Bimodal breathing was an important respiratory adaptation with evolutionary transition between aquatic and terrestrial habitats. • These animals use gills for water breathing and lungs for air breathing. • Trimodal Breathing is the ability of an organism to exchange respiratory gases through three surfaces i.e. skin, gills, and lungs.

	• Fundamental changes in the structure and function of the respiratory organs accompanied the transition from water to air breathing. • In terrestrial vertebrates i.e. reptiles, birds, and mammals, gills were replaced with lungs. • <u>Cutaneous Respiration</u> • It is the exchange of gases through skin. • Cutaneous exchange is most highly developed in frogs, toads, lungless salamanders and newts • Some more vertebrates such as some aquatic turtles, salamanders with lungs, snakes, fishes, and mammals (Bat). • These animals have lungs or gills, use cutaneous respiration to supplement gas exchange. • Amphibian skin has the simplest structure of all the major vertebrate respiratory organs. • In frogs, a uniform capillary network lies in a plane directly beneath the epidermis. • This vascular arrangement facilitates gas exchange between the capillary bed and the environment by both diffusion and convection. • A slimy mucous layer that keeps amphibian skin moist and protects against injury and helps in gas exchange. • Some amphibians obtain about 25% or more of their oxygen by this exchange, and the lungless salamanders carry out all of their gas exchange through the skin and buccal-pharyngeal region.
172	**Gill Respiration** • Gills are respiratory organs that have thin, moist epidermis over highly vascularized dermis for gaseous exchange. • Larval forms of a few fishes and amphibians have external gills projecting from their bodies. • Adult fishes have internal gills covered with operculum.

	• Below the operculum there are four gill arches, and each arch consists of many filaments. • A filament houses capillaries within lamellae where the direction of water flow opposes that of blood flow. • Below the operculum there are four gill arches, and each arch consists of many filaments. • A filament houses capillaries within lamellae where the direction of water flow opposes that of blood flow. • This counter current flow allows the fish to extract the maximal amount of oxygen from the water.
173	**Lung respiration in Amphibians and Reptiles** • Amphibian Respiration • As the tadpole grows, the gills disappear and lungs grow. • Amphibian lungs are simple saclike structures. • Internally lungs lack the complex spongy appearance unlike the lungs of birds and mammals. • Adult amphibians lack or have a reduced diaphragm, so breathing through the lungs is forced. • The lungs of most amphibians receive a large proportion of the total blood flow from the heart.  • Even though the amphibian ventricle is undivided, there is surprisingly little mixture of blood from the left and right atrial chambers within the single ventricle. • As a result, the lungs are perfused primarily with deoxygenated blood from the systemic tissues. • All reptiles breathe using lungs.

	• Aquatic turtles have developed more permeable skin, and some species have modified their cloaca to increase the area for gas exchange. • Because of this constraint, most Lizards and snakes are forced to hold their breath during intense runs. • In these reptiles the lungs are ventilated almost exclusively by the muscles of the chest wall. • Crocodiles and alligators have a large sheet of muscle below the lungs, called a diaphragm, which controls their breathing.
174	**Respiration in Birds** • Respiration in birds is complex and efficient. • It consists of lungs , 5-6 pairs of air sacs and air passages. • Trachea divides into right and left bronchi • Bird lungs have tunnel-like passages called parabronchi, which lead to air capillaries in which gas exchange occurs. • The end of each bronchus divides into small branches leading into air sacs. • These sacs are collapsible, and open and close as a result of muscle contractions around them. • The arrangement of air sacs make one-way flow possible. • Mechanism of Breathing • Inhaled air bypasses the lungs and enters the abdominal (posterior) air sacs. • It then passes through the lungs into the thoracic (anterior) air sacs. • Finally, air is exhaled from the thoracic air sacs. This whole process requires two complete breathing cycles. • Mechanism of Breathing • Inspiration and expiration result from alternate expansion and compression of the air sacs. • Inhaled air bypasses the lungs and enters the abdominal (posterior) air sacs. • It then moves to the thoracic (anterior) air sacs after passing through the lungs. • Finally, air is exhaled from the thoracic air sacs.

	• This whole process requires two complete breathing cycles. • Two cycles are required for the air to pass all the way through the system and out of the bird. 
175	**Human Respiratory system I** • Human respiratory system includes: • Air conducting portion 2. Gas exchange portion. • Air Conducting portion are the air passage ways that consist of nostrils, nasal cavities, pharynx, larynx, trachea, bronchi, bronchioles and alveolar ducts which ultimately leads into the alveolar sacs.   • As air enters through nostrils up to alveoli, air is filtered, moistened and conditioned. • In the pharynx the traffic of air and food is kept properly channeled by flap called epiglottis.

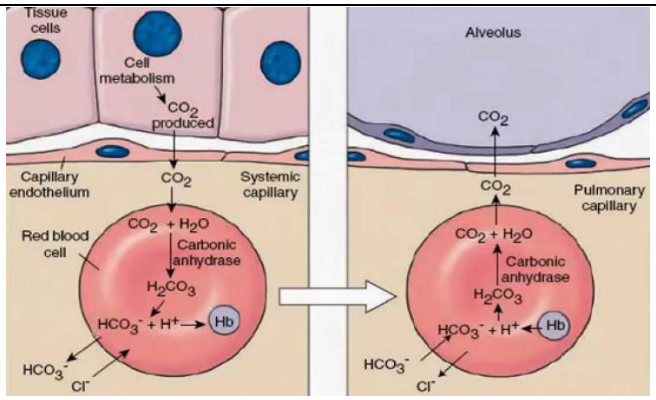
	• Alveoli are single layered endothelial walls. • Overlying the alveoli there is rich network of blood capillaries providing site for gaseous exchange. • The air passage ways are kept from collapsing by cartilage C-shaped rings in trachea and complete rings in bronchi. • The rings are progressively replaced by irregular cartilage plates. • The bronchioles totally lack cartilage but made of mainly circular smooth muscles.
176	**Human Respiratory system II** • The Lungs: Gas Exchange Portion of the Respiratory system: • Lungs are spongy closed sacs present in the chest cavity. • They are composed of millions of alveoli, a great amount of connective tissue and some muscles. • Lungs are covered by a thin layer of tough epithelium called visceral pleura. • A similar layer, parietal pleura lines the inner surface of the wall of the chest. • The cavity between these layers is called pleural cavity. • The chest cavity is bounded by spine, ribs and sternum. • The sides and floor are the intercostal muscles and muscular diaphragm. • Alveoli are the functional units of the lungs (gas exchange portion). Passive diffusion, driven by a partial pressure gradient, moves oxygen from the alveoli into the blood and moves carbon dioxide from the blood into the alveoli.
177	**Ventilation of Human Lungs** • Ventilation or Breathing has two phases: Inhalation (the intake of air) and Exhalation (the outflow of air)

	• These air movements result from the rhythmic increases and decreases in thoracic cavity volume. • Changes in thoracic volume lead to reversals in the pressure gradients between the lungs and the atmosphere; gases in the respiratory system follow these gradients. • The mechanism of inhalation operates in the following way: • Several sets of muscles, the main ones being the diaphragm and intercostal muscles, contract. • The intercostal muscles stretch from rib to rib, and when they contract, they pull the ribs closer together, enlarging the thoracic cavity. • The thoracic cavity further enlarges when the diaphragm contracts and flattens. • The increased size of the thoracic cavity causes pressure in the cavity to drop below the atmospheric pressure. • Air rushes into the lungs, and the lungs inflate. • During expiration or exhalation, air is expelled from the lungs in the following way: • The intercostal muscles and the diaphragm relax, allowing the thoracic cavity to return to its original, smaller size and increasing the pressure in the thoracic cavity. • Abdominal muscles contract, pushing the abdominal organs against the diaphragm, further increasing the pressure within the thoracic cavity. • The action in step 2 causes the elastic lungs to contract and compress the air in the alveoli. • With this compression, alveolar pressure becomes greater than atmospheric pressure, causing air to be expelled (exhaled) from the lungs.
178	• **Control of Respiration** • A breathing control center in the medulla sets the basic rhythm, of inhalations and exhalations. • Nerves from the medulla's control center send impulses to the breathing muscles-diaphragm and rib muscles.

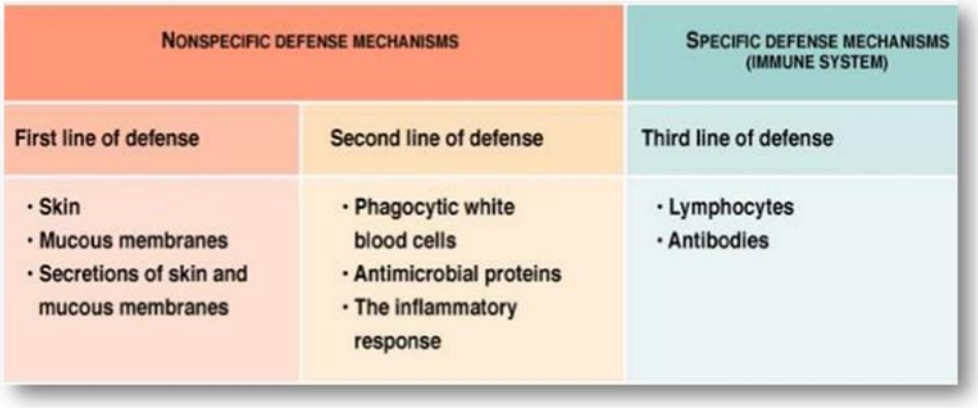
	• A person when at rest, these nerve impulses result in about 14 to 16 inhalations per minute. • Between two inhalations the muscles relax and the person exhales. • Sensors in the medulla detect changes in the pH of the blood and cerebrospinal fluid bathing the surface of the brain. • Sensors in major blood vessels detect changes in blood pH and send nerve impulses to the medulla. • In response, the medulla's control center alters the rate and depth of breathing, increasing both if CO_2 levels rise or decreasing both if CO_2 levels fall. • Other sensors in the aorta and carotid arteries signal the medulla to increase the breathing rate when O_2 levels in the blood become very low.
179	• **Transport of Respiratory Gas 'Oxygen'** • As we know that oxygen must be transported from the environment to each and every cell of an animal's body. • Various systems (e.g., tracheae, cutaneous exchange, gills, lungs) help accomplish this transport. • As animals became larger and acquired higher metabolic rates, simple diffusion became increasingly inadequate as a means of delivering oxygen to the tissues • In general, more active animals have an increased demand for oxygen. • Water-based body fluid does not guarantee internal transport of sufficient oxygen to meet this increased demand. • The reason is the low solubility of oxygen in water-based body fluids. • Thus, fluid-borne respiratory pigments reversibly binding large quantities of oxygen evolved in most phyla. • Water-based body fluid does not guarantee internal transport of sufficient oxygen to meet this increased demand. • The reason is the low solubility of oxygen in water-based body fluids. • Thus, fluid-borne respiratory pigments reversibly binding large quantities of oxygen evolved in most phyla.
180	**Respiratory Pigments I**

	• Hemoglobin is a reddish pigment that contains iron as the oxygen-binding metal. • It is the most common respiratory pigment in animals, being found in a variety of invertebrates (e.g., platyhelminths, nemerteans, nematodes, annelids, crustaceans, some insects, and molluscs. • All vertebrates with the exception of a few fishes. • Icefish (Channichthyidae) are the only known vertebrates that lack hemoglobin in their blood. • This wide distribution suggests that hemoglobin evolved very early in the history of animal life. • Hemoglobin may be carried within red blood cells (erythrocytes or simply dissolved in the blood or coelomic fluid. • $Hb + O_2 = HbO_2$ (Under high O_2 concentration) (in lungs) • $HbO_2 = Hb + O_2$ • (Under high O_2 concentration) (in tissues) • Factor that affect oxygen combining capacity of Hb: a) Carbon Dioxide b) Temperature c) pH of blood d) 1% CO in blood cause death. c) Hb conc. increases at high altitudes. • Human fetuses have high conc. of Hb.
181	**Respiratory Pigments II** • Hemocyanin is the most commonly occurring respiratory pigment in molluscs and certain crustaceans.

	• Hemocyanin contains metallic copper, has a bluish color when oxygenated, and always occurs dissolved in hemolymph. • Unlike most hemoglobin, hemocyanin tends to release oxygen easily and to provide a ready source of oxygen to the tissues. • Hemerythrin contains iron and is pink when oxygenated. • It is in nucleated cells, rather than free in body fluids or hemolymph. • A few brachiopods and some polychaetes have hemerythrin. • Chlorocruorin also contains iron but is green when associated with low oxygen concentrations and bright red when associated with high oxygen concentrations. • Chlorocruorin occurs in several families of polychaete worms.
182	**Transport of Respiratory Gas (Carbon Dioxide)** • Carbon dioxide produced within the respiring cells diffuse into blood, then transported to lungs from where exhaled. • CO_2 is transported in the following three states: a. As carboxyhemoglobin (about 20% of CO_2) combines with amino group of Hb. b. CO_2 as such (5%) c. As bicarbonates (75%) • Bicarbonates are formed from CO_2 and H_2O in the tissues by the following reaction and Process. • Reaction in Tissues: CO_2 and H_2O combine to form Carbonic acid. • This reaction is catalyzed by carbonic anhydrase present in RBCs. • Carbonic acid then ionizes into HCO^{-3} and H^{+} • Most of CO_2 is transported as HCO^{-3} ions from the tissues to the lungs. • Reactions in Lung Capillaries: • $HCO^{-3} + H^{+} = H_2CO_3$ • H_2CO_3----------H_2O+CO_2 • CO_2 thus formed passes out of the capillaries into the alveoli from where it is exhaled.

	
183,184	• **Summary of chapter I** • Any system that supplies the body tissues with oxygen, nutrients, hormones, and removes waste products is called transport system or circulatory system. • The two basic types of circulatory systems are open and closed. Open systems generally circulate hemolymph, and closed systems circulate blood. • Blood is a type of connective tissue made up of blood cells (red blood cells and white blood cells), plasma, and platelets. • The heart pumps blood through a series of vessels in the following order: arteries, arterioles, capillaries, venules, veins, and back to the heart. • The action of the heart consists of cyclic contraction (systole) and relaxation (diastole). Systolic contraction generates blood pressure that forces blood through the closed system of vessels. • The lymphatic system consists of one-way vessels that help return fluids and proteins to the circulatory system. • Animals that respire aerobically need a constant supply of oxygen. • The process of acquiring oxygen and eliminating carbon dioxide is called external respiration. • The exchange of oxygen and carbon dioxide occurs across respiratory surfaces. • Such surfaces include gills, cutaneous surfaces, and lungs. • The air-conducting portion of the respiratory system of air-breathing vertebrates moves air into (inhalation) and out of (exhalation) this system. This process of air movement is called ventilation.

	• Oxygen and carbon dioxide diffuse from areas of higher concentration to areas of lower concentration. • Once in the blood, oxygen diffuses into red blood cells and binds to hemoglobin for transport to the tissues. • Carbon dioxide is transported bound to hemoglobin, as well as in the form of the bicarbonate ion and carbonic acid. • Respiratory pigments are organic compounds that have either metallic copper or iron that binds oxygen. • Examples include hemoglobin, hemocyanin, hemerythrin, and chlorocruorin.
185	• **Introduction of Immune System** • Animals are constantly under attack by infectious agents that cause disease. • For a pathogen, an animal body is a nearly ideal habitat. • The host offers a ready source of nutrients, a protected setting for growth and reproduction. • Hosts also provide means of transport to new hosts and environments. • Having this opportunity, pathogens, mostly viruses, bacteria, protists, and fungi, infect a wide range of animals, including humans. • If not eliminated they may deal with abnormal cells and develop into cancer. • To counter these threats animals have evolved defense systems. • Additional responses to infection take many forms, including proteins that punch holes in bacterial membranes or block viruses from entering body cells. • These and other defenses make up an immune system, which enables an animal to avoid or limit many infections.

	
186	**Non Specific defense mechanism** • Nonspecific mechanisms provide general barriers to infection. • This includes two lines of defense, which an invader encounters in sequence. • <u>First line of defense is external that includes:</u> a) skin b) mucous membranes and their secretions. • <u>Second line of defense is internal that include</u>: a) phagocytic WBCs b) antimicrobial proteins c) inflammatory response • An invading microbe must penetrate external barrier formed by skin and mucous membranes. • Skin covers the surface and the mucous membrane lines the openings of the animal. <u>Skin</u> is a barrier that cannot normally be penetrated by bacteria or viruses. <u>Mucous membrane</u> lines the digestive, respiratory and genitourinary tract bar the entry of microbes. • Secretions from oil and skin glands prevent bacteria from colonizing. • Saliva, tears and mucous secretions bathe the surface washing away many invaders.

	• Microbes entering the upper respiratory system are often caught in mucus are either swallowed or expelled.
	• Bone Marrow is a semi-solid tissue which is found within the spongy or portions of bones. • In birds and mammals, bone marrow is the primary site of new blood cell production or hematopoiesis. • It is composed of hematopoietic cells and adipose tissue. • In adult humans, bone marrow is primarily located in the ribs, vertebrae, sternum, and bones of the pelvis. • Human marrow produces approximately 500 billion blood cells per day. • The red bone marrow is a key element of the lymphatic system. • In adult humans, bone marrow is primarily located in the ribs, vertebrae, sternum, and bones of the pelvis. • Human marrow produces approximately 500 billion blood cells per day. • The red bone marrow is a key element of the lymphatic system.
187	**Specific defense mechanism (Immune System)** • Lymphocyte is a type of white blood cell (leukocyte) that is of fundamental importance in the immune system. • Lymphocytes are the cells that determine the specificity of the immune response to pathgens and other foreign substances. • There are three main types of lymphocytes: a) T cells, b) B cells, and c) Natural killer cells. • T cell is a type of lymphocyte which mature in the thymus . • T cells are different from other lymphocytes by the presence of a T-cell receptor on the cell surface.

	• B cells are responsible for generating antibodies to specific antigens, which they bind via B cell receptors (BCR). • Natural killer cells is a component of innate immune system. • NK cells play a major role in the host-rejection of both tumors and virally infected cells. • Antigens: The foreign substance that elicits the immune response is called an antigen. • The immune system responds to an antigen by producing specific proteins called antibodies. • The antigens include molecules belonging to viruses, bacteria, fungi and parasitic worms. • Each antigen has a unique molecular shape and stimulates the production of antibodies. • Antigens are bound to antibodies through weak and non covalent interactions such as electrostatic interactions, hydrogen bonds, Van der Waals forces, and hydrophobic interactions.
188	**Acquired Immunity** • Acquired immunity: Acquired or adaptive immunity is the body's third line of defense. • This is protection against specific types of pathogens. • Acquired immunity may be either natural or artificial in nature. • Both natural and artificial immunity have active and passive components. • Active immunity . the immunity which results from the production of antibodies by the immune system in response to the presence of an antigen. • Active immunity comes from exposure to a pathogen. • E. Jenner, English physician and scientist was the pioneer of smallpox vaccine, the world's first vaccine. • Vaccine contains a small amount of the disease germ (virus or bacteria) or parts of the germ. Examples are the measles virus, whooping cough bacteria and tetanus toxoid. • Surface markers on the pathogen surface act as antigens, which are binding sites for antibodies.

	• Antibodies are Y-shaped protein molecules, which can exist on their own or attach to the membrane of special cells. • The body doesn't keep a store of antibodies on hand to take down an infection immediately. • Naturally acquired active immunity occurs when the person is exposed to a live pathogen, develops the disease, and becomes immune as a result of the primary immune response. • Once a microbe penetrates the body's skin, mucous membranes, or other primary defenses, it interacts with the immune system. • Naturally acquired active immunity occurs when the person is exposed to a live pathogen, develops the disease, and becomes immune as a result of the primary immune response. • Once a microbe penetrates the body's skin, mucous membranes, or other primary defenses, it interacts with the immune system. • An example of natural passive immunity is a baby's protection against certain infections by getting antibodies through colostrum or breast milk. • An example of artificial passive immunity is getting an injection of antisera, which is a suspension of antibody particles. • Artificially acquired active immunity can be induced by a vaccine, a substance that contains the antigen.
189	**Vaccines** • Vaccine is a substance used to stimulate the production of antibodies and provide immunity against one or several diseases. • Vaccines are prepared from the causative agent of a disease or its products. • It may be a synthetic substitute, treated to act as an antigen without inducing the disease. • "there is no vaccine against the virus“. • Vaccine is the suspension of weakened or killed microorganisms or toxins or antibodies or lymphocytes administered to prevent disease. • There are 4 main types of vaccines: • Live-attenuated vaccines.

	• Inactivated vaccines. • Subunit, recombinant, polysaccharide, and conjugate vaccines. • Toxoid vaccines. • The first inactivated polio vaccine (IPV) was produced by Salk using virus grown on monkey kidney cells and inactivated with formalin. • Attenuated or Weakened • Inactivated (Killing by formaline) • measles, mumps, and rubella (MMR)
193	• **Passive Immunity** • Passive immunity is conferred from outside the body, so it doesn't require exposure to an infectious agent or its antigen. • There is no delay in the action of passive immunity. Its response to an infectious agent is immediate. Passive immunity is not as long-lasting as active immunityAn example of natural passive immunity is a baby's protection against certain infections by getting antibodies through colostrum or breast milk. • An example of artificial passive immunity is getting an injection of antisera, which is a suspension of antibody particles. • Artificially-acquired passive immunity is an immediate, but short-term immunization provided by the injection of antibodies, such as gamma globulin, that are not produced by the recipient's cells. • These antibodies are developed in another individual or animal and then injected into another individual. 